

Data Analytics Foundations

Chapter 1 – Hands on Data Analytics for Everyone

August 27, 2026

北师香港浸会大学
BEIJING NORMAL-HONG KONG BAPTIST UNIVERSITY

Contents

- **Background**
- **Data Analytics Project Lifecycle**
  - **Example of Data Analytics Project**
  - **Project Understanding Phase**
  - **Data Understanding Phase**
- **Data Format**
- **Data Types**
- **Lab: Workflow in KNIME (Data Import Export)**
- **Lab: Excel and CSV tables**
- **Review Questions**
- **APPENDIX: Introduction to KNIME**

What is Data Science?



Data science is a multi-disciplinary field that uses scientific methods, processes, algorithms and systems to **extract knowledge and insights** from structured and unstructured data.

Think of it as a blend of skills that allows you to:

- **Collect** large amounts of information (data).
- **Clean** and organize it.
- **Analyze** it to find hidden patterns, trends, and answers.
- **Communicate** those findings in a way that helps people or companies make better decisions.

What is Data Science? From Data To Knowledge



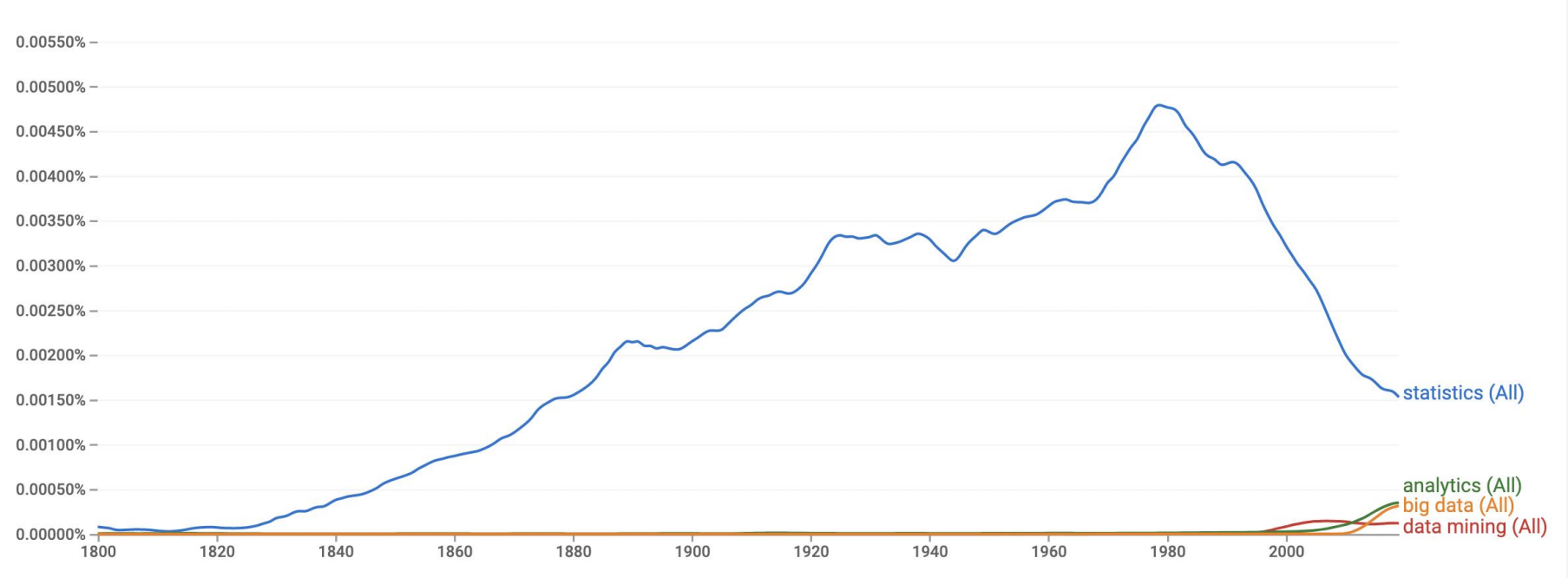
Data

- refer to single instances (single objects, people, events, points in time, etc.)
- describe individual properties
- are often available in large amounts (databases, archives)
- are often easy to collect or to obtain (e.g., scanner cashiers in supermarkets, Internet)
- do not allow us to make predictions or forecasts

Knowledge

- refers to *classes* of instances (*sets* of objects, people, events, points in time, etc.)
- describes general patterns, structures, laws, principles, etc.
- consists of as few statements as possible
- is often difficult and time consuming to find or to obtain (e.g., natural laws, education)
- allows us to make predictions and forecasts

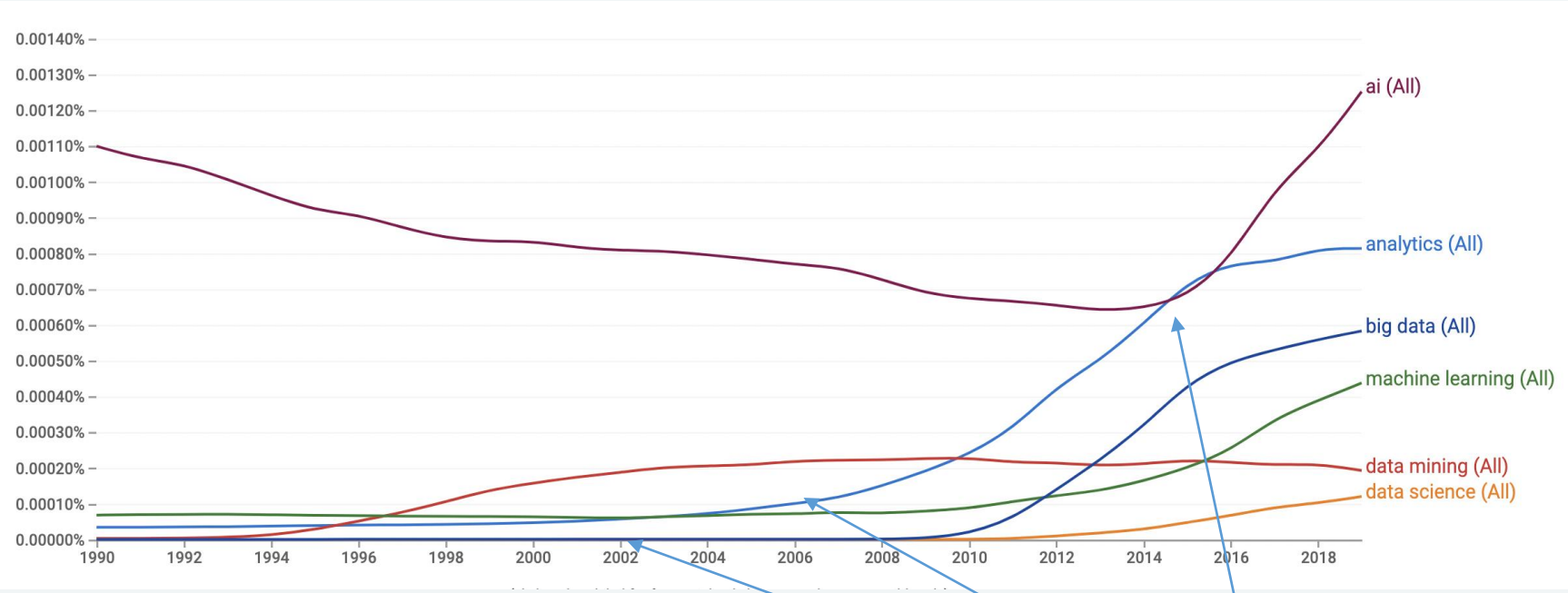
The Era of Statistics (1800-2000)

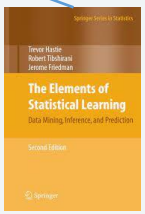


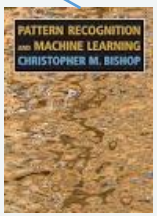
Source: Google Book Ngram viewer (case insensitive search)

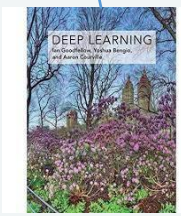
- Analytics and big data appearing and increasing around 2010
- data mining appears in late 1990s
- Statistics reached peak in 1980 and it is decreasing since then.

The Era of Data Science









- Data mining surges in 1996
- AI re-surges in 2015 after declining
- Big Data, analytics, machine learning and data science all surges in between 2008-2015

What the future holds?

- 1900s - Statistics
- 1960s “Data Mining” is a bad activity (data “dredging” meaning)
- 1990 - “Data Mining” is good, surges in 1996 and peak in 2003
- 2010s - Business/Data/Predictive Analytics, Machine Learning and Data Science
  - 2014 - Big Data / Deep Learning
  - 2018 – A.I.
- -- 2030 ?

Software engineering | Robotics | Liberal Arts



Main Industries for Data Analytics According to Experts

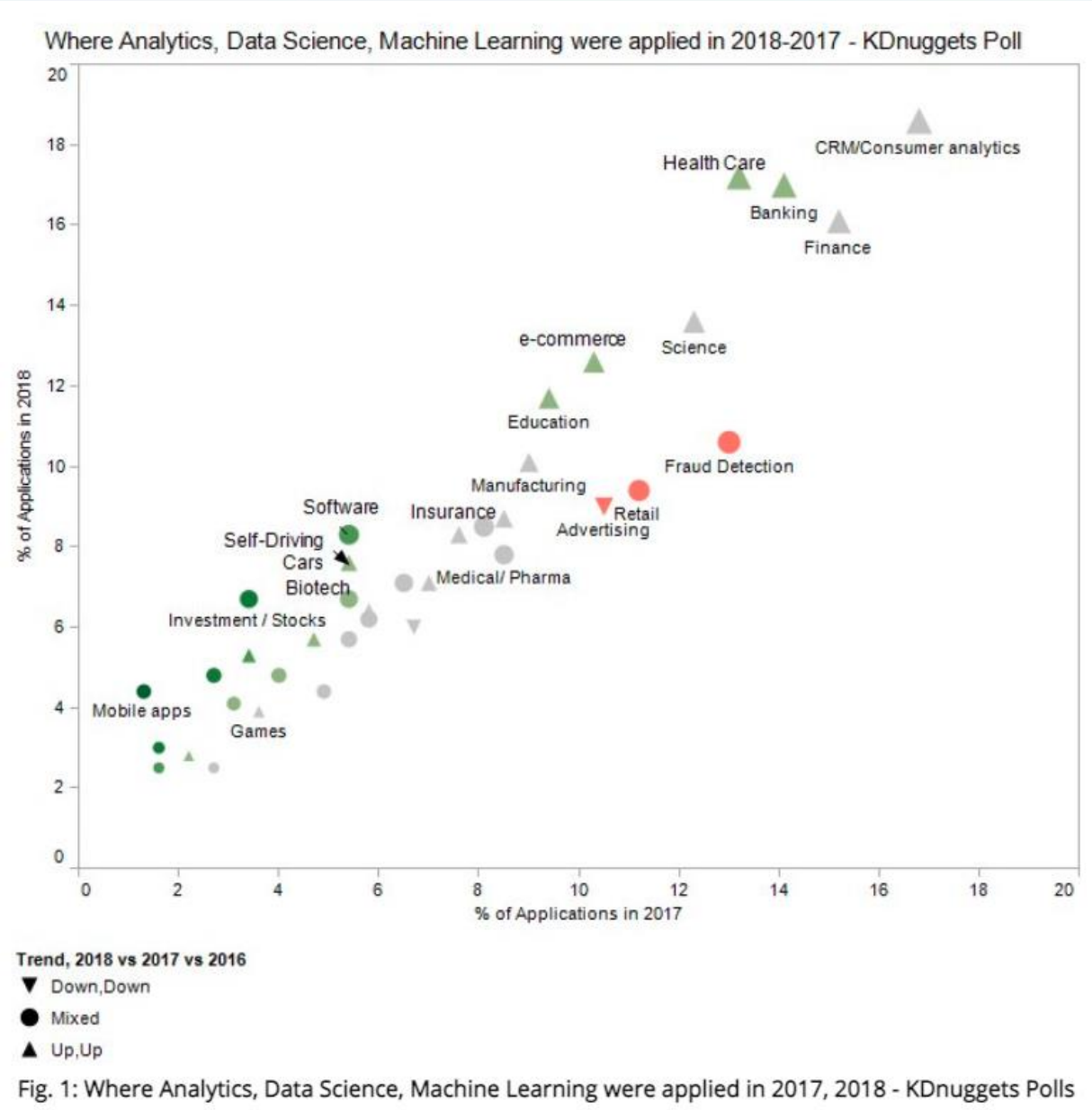


Fig. 1: Where Analytics, Data Science, Machine Learning were applied in 2017, 2018 - KDnuggets Polls

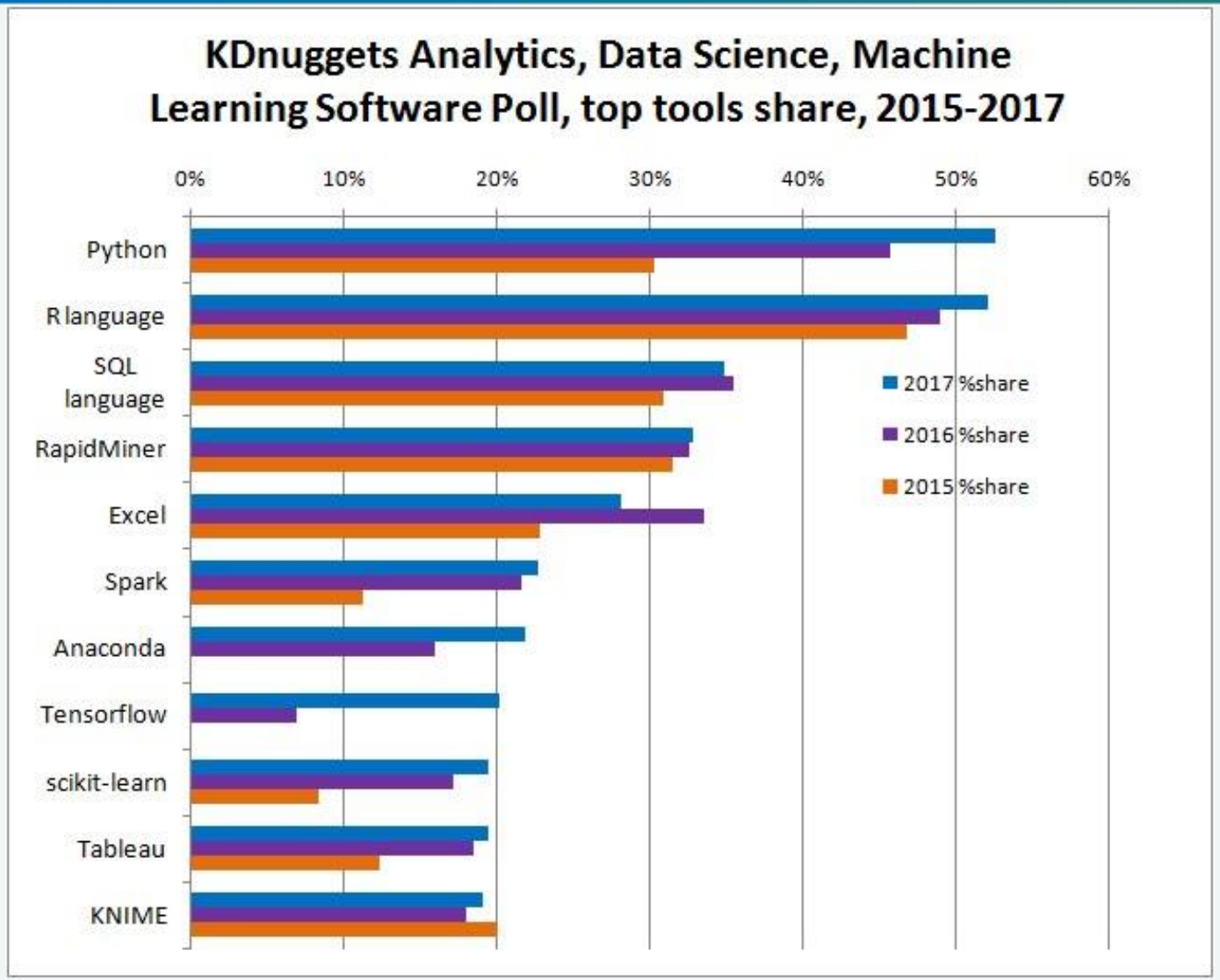
Source: Kdnuggets
https://www.kdnuggets.com/2019/03/poll-analytics-data-science-ml-applied-2018.html

Readers of KDnuggets were asked:
What are the Industries / Fields where you applied Analytics, Data Science, Machine Learning in 2018?

The top 5 industries were:

1. **CRM/Consumer analytics**, 18.6% (number 1 for the last 5 years)
2. **Health care**, 17.2% (jump to number 2 from number 4)
3. **Banking**, 17.0%
4. **Finance**, 16.1%
5. **Science**, 13.6%

Main Software/Languages for Data Analytics



Source: Kdnuggets
https://www.kdnuggets.com/2017/05/poll-analytics-data-science-machine-learning-software-leaders.html

Some Data Science Roles in the Industry



- Data Analyst
- (Big) Data Engineer
- Data Scientist
- Machine Learning Researcher
- Data Science Manager/Director
- Chief Data Officer

中国十大高薪行业

（找工作必备宝典）

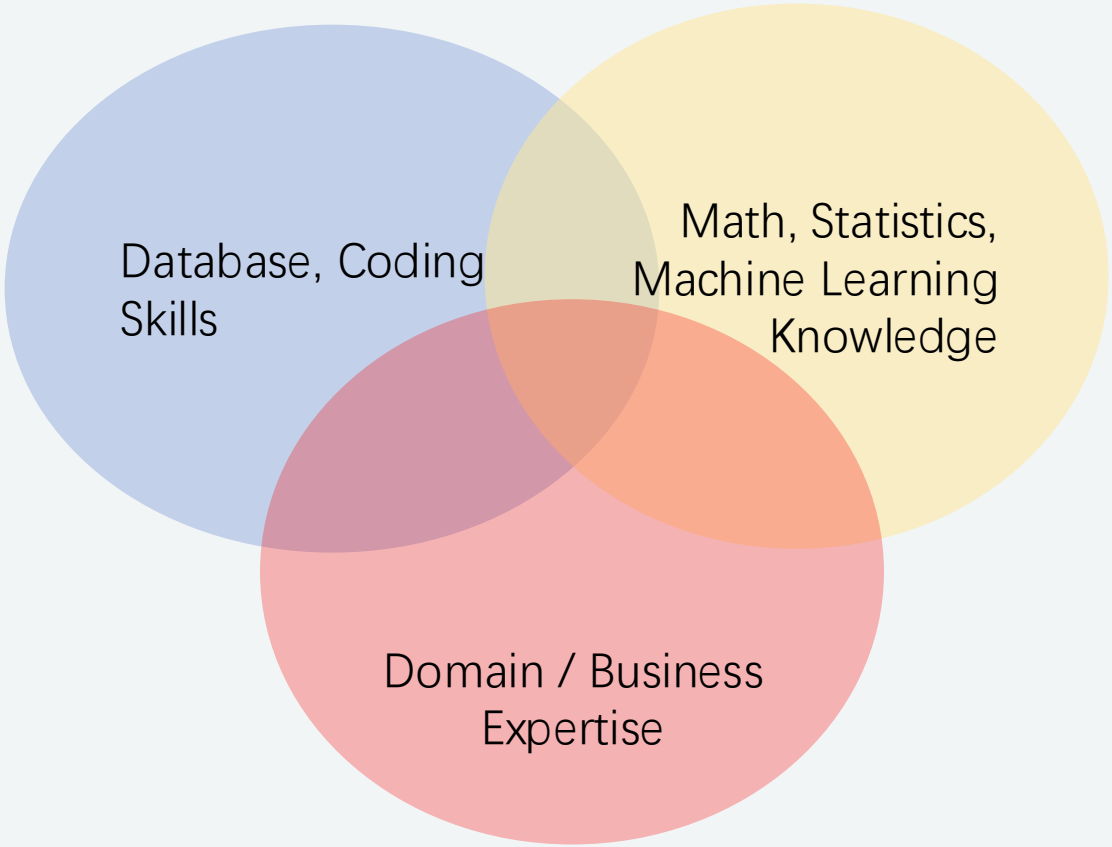
*排名为该行业综合薪资水平个别行业个例不纳入参考范围

注意高薪诈骗

*以下仅为参考,具体请查看各公司公开招聘信息



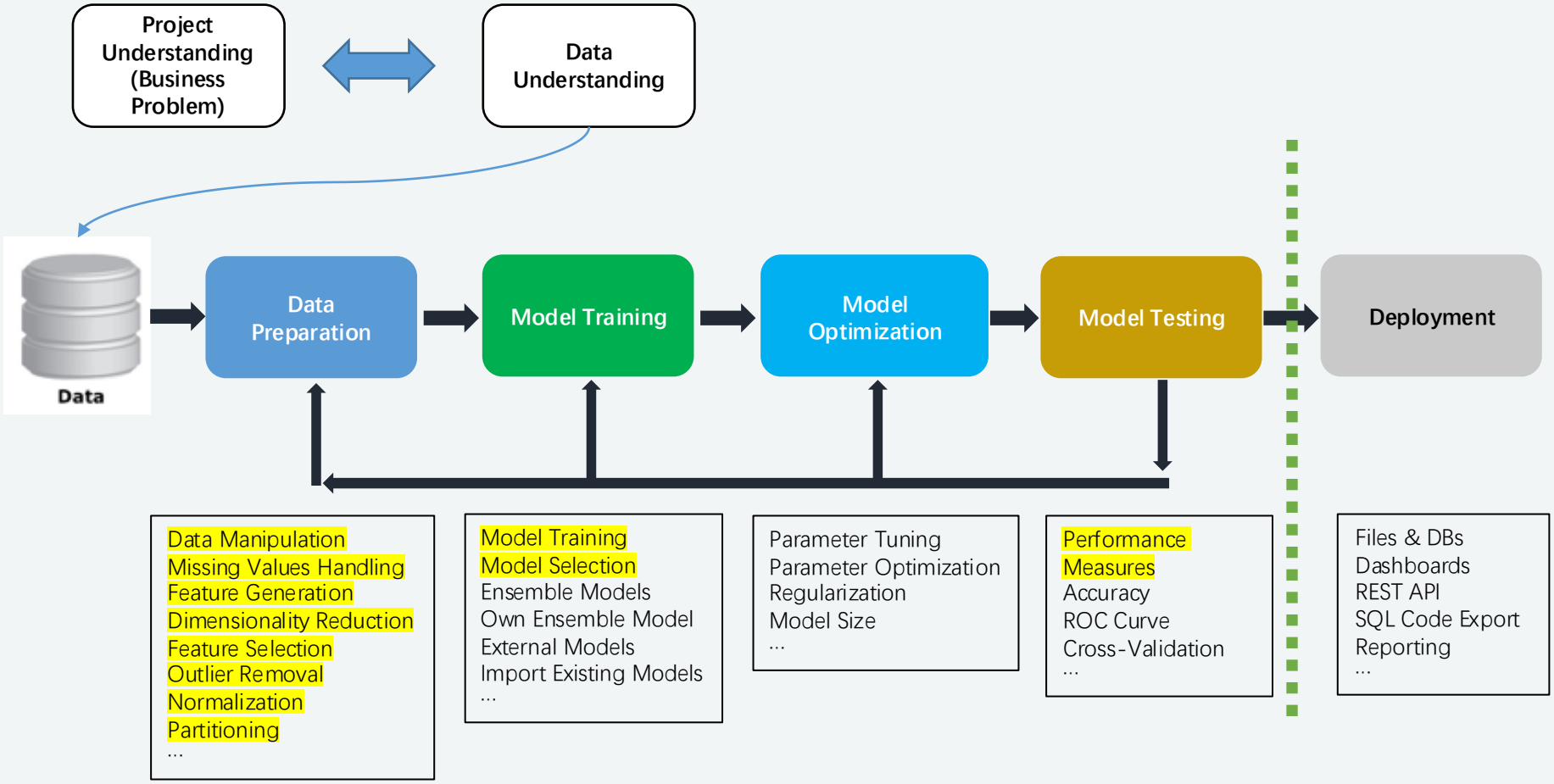
Data Science Skills



Contents

- Background
- Data Analytics Project Lifecycle
  - Example of Data Analytics Project
  - Project Understanding Phase
  - Data Understanding Phase
- Data Format
- Data Types
- Lab: Workflow in KNIME (Data Import Export)
- Lab: Excel and CSV tables
- Review Questions
- APPENDIX: Introduction to KNIME

Process Flow of a Data Science Project



Contents

- Background
- Data Analytics Project Lifecycle
  - Example of Data Analytics Project
  - Project Understanding Phase
  - Data Understanding Phase
- Data Format
- Data Types
- Lab: Workflow in KNIME (Data Import Export)
- Lab: Excel and CSV tables
- Review Questions
- APPENDIX: Introduction to KNIME

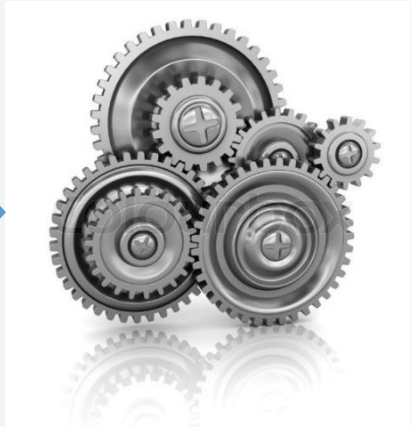
Example 1: Customer Segmentation

Business question: Which groups of customers am I serving?



CRM System
Data about your customer

- Demographics
- Behavior
- Revenues



Model



- Churn Prediction
- Upselling Likelihood
- Product Propensity /NBO
- Campaign Management
- Customer Segmentation
- …

Example 1



- Dataset from a hypothetical supermarket chain
  - Customers
  - Products
  - Purchases

- Three tasks
  - Divide customers into different groups according to their purchase behaviour
  - Identify connections between products to implement cross-selling campaigns
  - Helping design a marketing campaign to increase purchases

- Two approaches
  - Naive approach lead by common sense
  - Sound approach using data analytics techniques

Example 1: Data Understanding and Pattern Finding



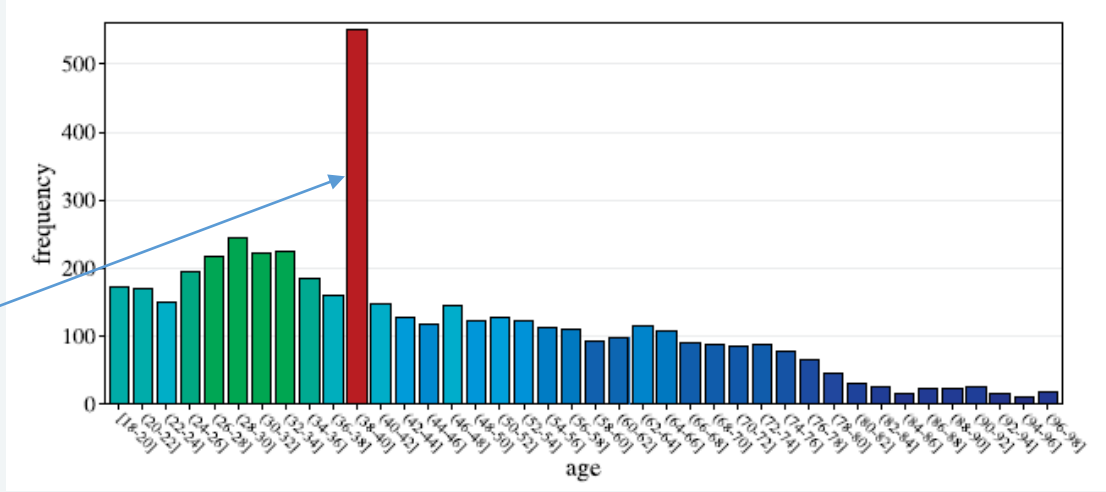
- A Naive Approach
  - Compute the age of each customer
  - Run an algorithm (cluster analysis) to group customers based on their age and expenditure

  ➢ No interesting relations highlighted in this analysis

Cluster ID	Age (Cluster Average)	Total Purchase (Cluster Average)
1	46.5	€ 1,922.07
2	39.4	€ 11,162.20
3	39.1	€ 7,279.59
4	46.3	€ 419.23
5	39.0	€ 4,459.30

Example 1: Data Understanding and Pattern Finding

- Sound Approach – Check Data
  - Check customer names, employment if match expectations
  - Check and add constraints to numeric attributes (e.g. Age must be between 18-100)
  - Look for misleading information (e.g. a missing birthdate was automatically set by default set to 1970, these cases must be handled)



- Use meaningful measures
  - Use average basket price as estimator for the value of a customer
  - Use average number of purchases per month as further estimator
- Apply the clustering algorithm
  - Apply normalization of measures to average magnitudes and compare across different features

Cluster	Age	Avg. cart price	Avg. purchase/month
1	75.3	€ 19.-	5.6
2	42.1	€ 78.-	7.8
3	38.1	€ 112.-	9.3
4	30.6	€ 16.-	4.8
5	44.7	€ 45.-	3.7

Example 1: Data Understanding and Pattern Finding



Cluster	Age	Avg. cart price	Avg. purchase/month
1	75.3	$ 19.-	5.6
2	42.1	$ 78.-	7.8
3	38.1	$ 112.-	9.3
4	30.6	$ 16.-	4.8
5	44.7	$ 45.-	3.7

Interpretation:

- Cluster (#1) of older customers have a relatively small average basket price
- Cluster (#4) of younger shoppers also purchasing smaller baskets.
- The middle-aged group varies wildly in price, however.

Assumption about family status:

- People with families will likely buy more products and hence combine more products into more expensive baskets, which seems to explain the difference between clusters #2/#3 and cluster #5.

Further Analysis:

- Consider product categories
- Rules match with well-known facts
- Monitor combinations on regular basis

Association Analysis:

tomatoes ← capers, pasta [support=0.007, confidence=32%]

tomatoes ← apples [support=0.013, confidence=22%]

Example 1: Targeted Marketing from Customer Segmentation

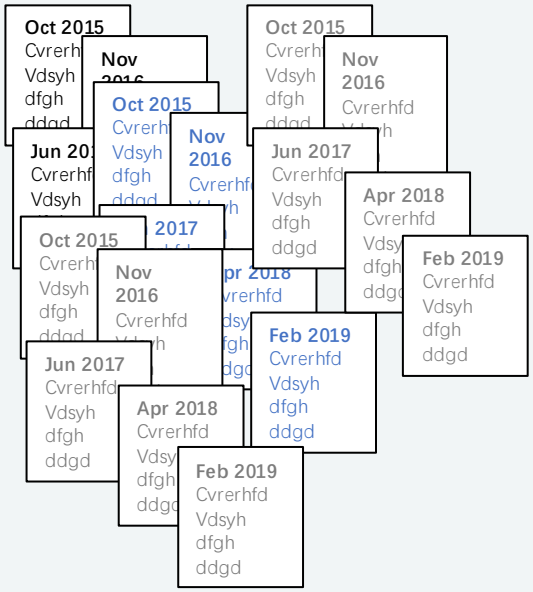
After the analysis, the data analyst, marketing and campaign specialists meet to develop strategies to foster the customer groups which bring in the largest chunk of revenue and develop the ones which seem to be underrepresented

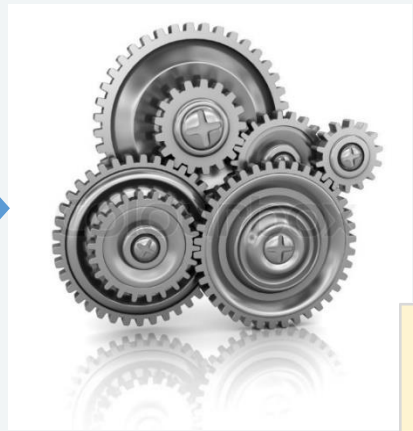
- Discount program targeting the basket size of customers. For example:
  - discount of 1$ on food purchase targeting cluster #1
  - discount of 10$ on electronics purchase targeting cluster #3
- Predict the need and recommend product to its existing customers. For example recommend capers and pasta for those who purchased tomato

Example 2: Risk Assessment

Business question: Is this person going to repay the loan?

Customer History





Model

Risk Prognosis

- High Risk
- Low Risk
- High Risk
- Very High Risk
- Very Low Risk
- Medium Risk
- ...

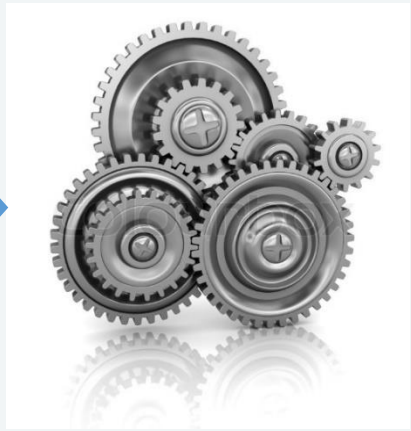


Example 3: Demand Prediction

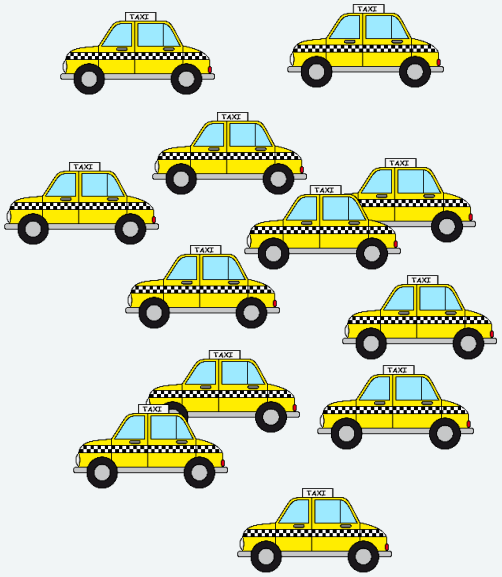
Business questions:

- How many taxis do I need in NYC on Wednesday at noon?
- Or how many kW will be required tomorrow at 6am in London?
- Or how many customers will come tonight to my restaurant?



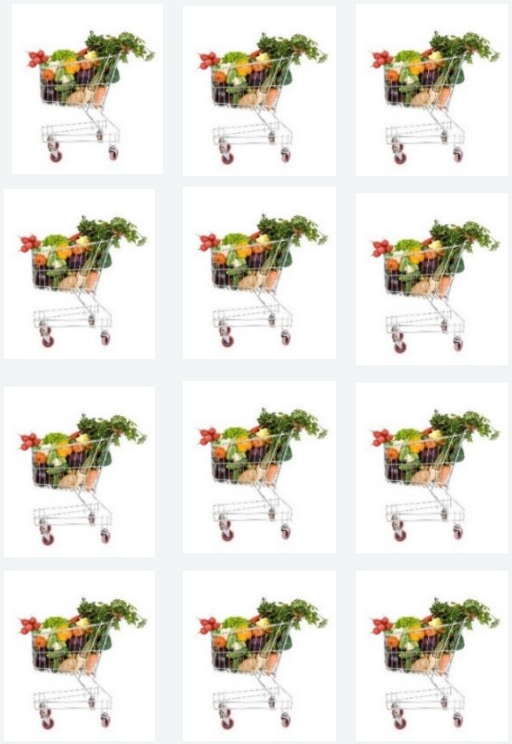


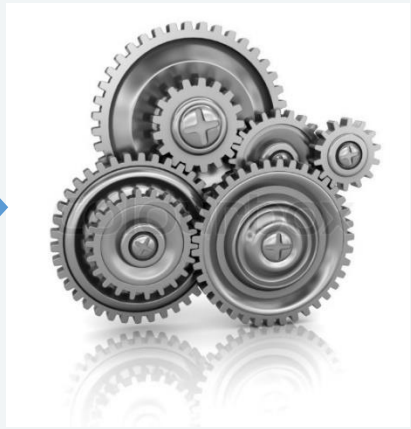
Model



Example 4: Recommendation Engines / Market Basket Analysis

Business question: People who bought this item were often interested in this other items.





Model

Recommendation

IF  ➔ 

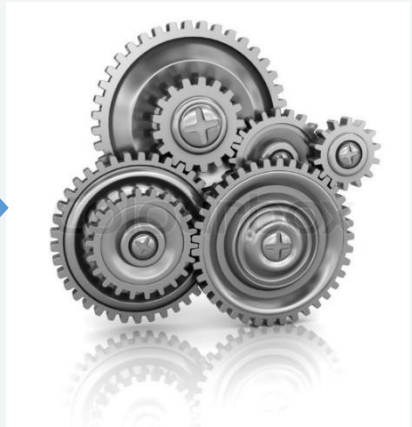
Example 5: Fraud Detection

- Business question: Is this transaction legitimate or is it a fraud?

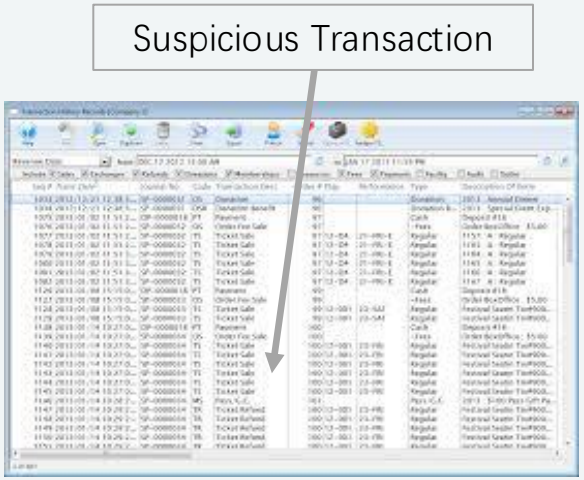


Transactions

- Trx 1
- Trx 2
- Trx 3
- Trx 4
- Trx 5
- Trx 6
- …



Model



Example 6: Sentiment Analysis

- Business question: how can I know what people are thinking?



Samsung

Samsung Galaxy S7 Edge G935A 32GB Unlocked - Gold Platinum

★★★★☆ 125 customer reviews | 606 answered questions

★★★★★ **Beautiful phone from a wonderful seller!**

By Atheism-the-Buddhist-Way on May 29, 2017

Color: Gold | **Verified Purchase**

This practically new beautiful phone well exceeded my expectations!



★☆☆☆☆ **One Star**

By Christopher white on August 3, 2016

Color: Black Onyx | **Verified Purchase**

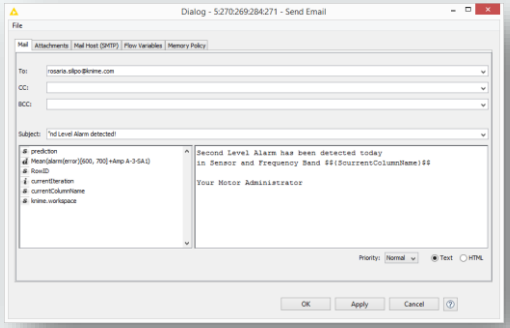
Very bad experience



Example 7: Anomaly Detection

Business question: Can I predicting mechanical failure before it happens?







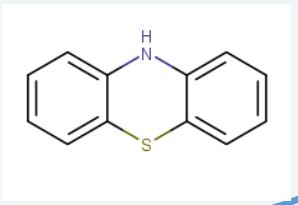


Only some Spectral Time Series show the break down

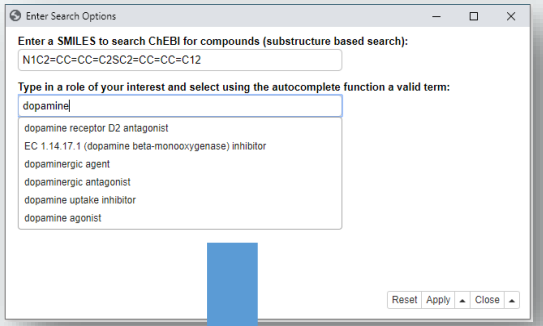
Example 8: Compound Search (Analytical Chemistry)

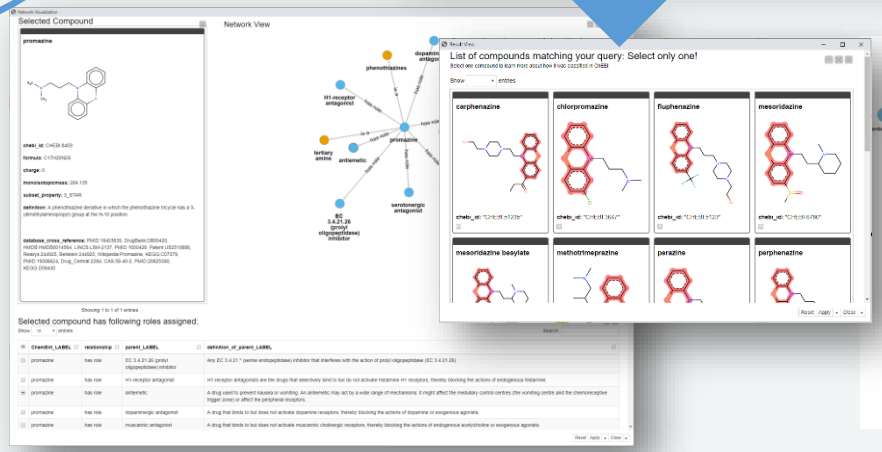
Questions: Improve product quality, material yield and properties of the material

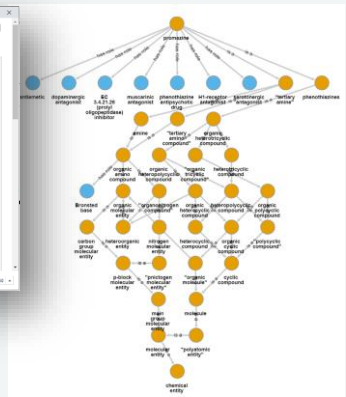
Are there other compounds having this substructure and being a dopaminergic antagonist?











Contents

- Background
- Data Analytics Project Lifecycle
  - Example of Data Analytics Project
  - Project Understanding Phase
  - Data Understanding Phase
- Data Format
- Data Types
- Lab: Workflow in KNIME (Data Import Export)
- Lab: Excel and CSV tables
- Review Questions
- APPENDIX: Introduction to KNIME

Determine the Project Objective



- What is the primary objective?
- What are the criteria for success?

Difficulty: The project owner & the analyst *speak different languages*

Determine the Project Objective



Example:

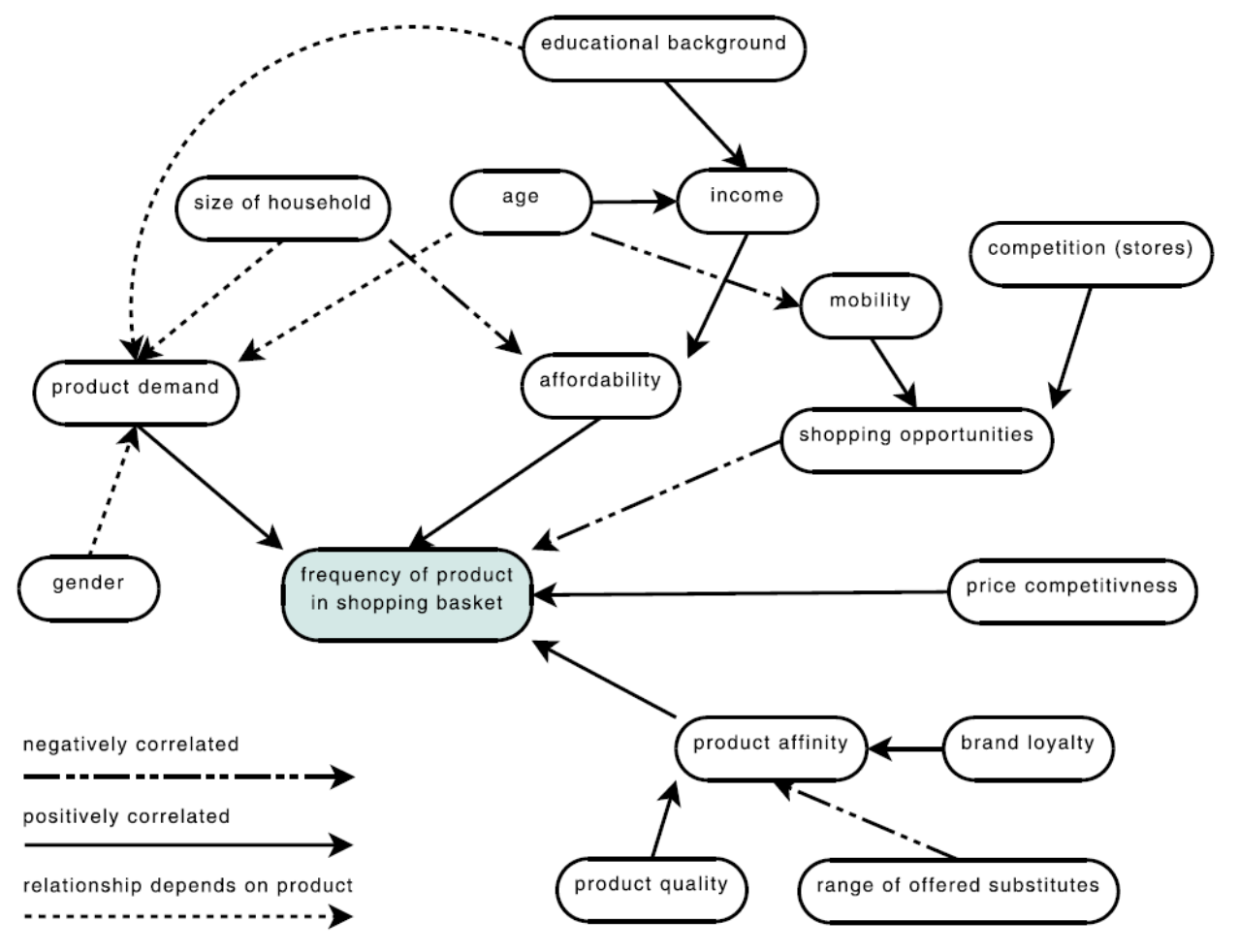
Objective	Increase revenues (per campaign and/or per customer) in direct mailing campaigns by personalized offer and individual customer selection
Deliverable	Software that automatically selects a specified number of customers from the database to whom the mailing shall be sent, runtime max. half-day for database of current size
Success criteria	Improve order rate by 5% or total revenues by 5%, measured within 4 weeks after mailing was sent, compared to rate of last 3 mailings

- Once the solution is identified
  - Explore advantages & disadvantages
- Is the goal
  - Precise enough?
  - Actionable?

Cognitive Maps

Sketching tool:

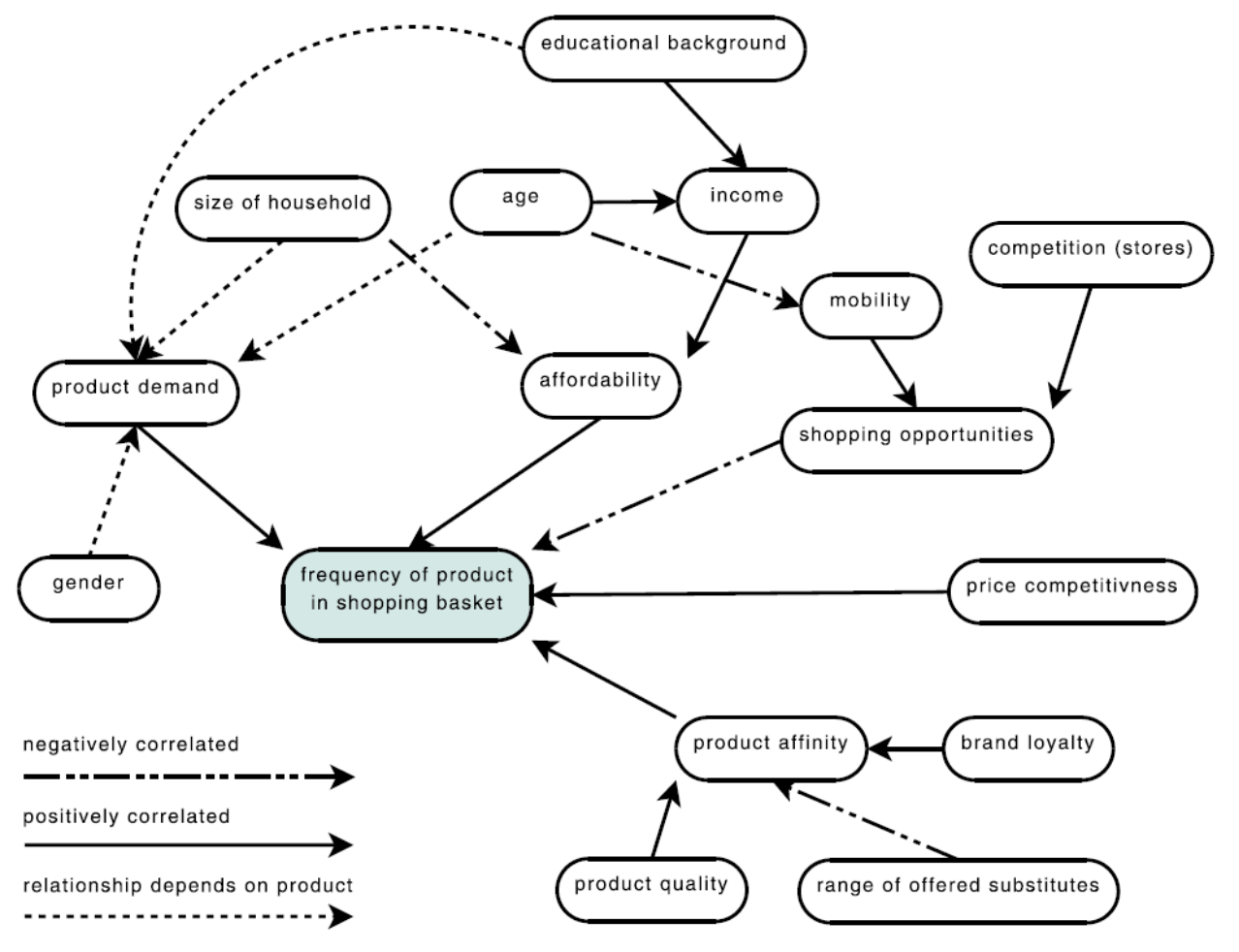
- Beliefs
- Experiences
- Known factors
- How factors influence each other



Cognitive Maps

How often will a certain product be found in a basket?

- Directly influenced by factors around it
  - E.g., affordability
- Indirectly influenced by other factors
  - E.g., size of household
- Postive or negative correlation



Assess the Situation



- Will this be a successful data analysis project?

Examine the following:

- **Requirements and constraints**
  - Model requirements (e.g., explanatory model)
  - Ethical, political, and legal issues (e.g., must exclude gender, race, and/or age)
  - Technical constrains
- **Assumptions**
  - Representativeness (the sample represents the whole population)
  - Informativeness (influencing factors should be included in the model)
  - Good data quality
  - Presence of external factors

Determine Analysis Goals



Select models and techniques with the following properties

- **Interpretability**
  - The model can be understood / interpreted
- **Reproducibility / stability**
  - Similar model performance every time the analysis is carried out
- **Model flexibility / adequacy**
  - The model can adapt to more complicated situations
- **Runtime**
  - Strict runtime requirements may limit computationally intensive approaches
- **Interestingness / use of expert knowledge**
  - Experts may already know the findings from the analysis

Contents

- Background
- **Data Analytics Project Lifecycle**
  - Example of Data Analytics Project
  - Project Understanding Phase
  - **Data Understanding Phase**
- Data Format
- Data Types
- Lab: Workflow in KNIME (Data Import Export)
- Lab: Excel and CSV tables
- Review Questions
- APPENDIX: Introduction to KNIME

Problems with Data and Data Cleaning



Getting the data is not always easy:

- Different resources: flat files, different databases, excel spreadsheets, ...
- Integration is cumbersome: Missing/not unique IDs, wrong entries, ...
- Sometimes also privacy concerns (not all data in one location)

Data needs to be transformed:

- Type conversions
- Missing value correction/clean up/imputation
- Generation of new values (e.g. convert year of birth into age)

Problems with Data and Data Cleaning



In Example 1 we have 3 files:

- customers,
- products,
- shopping baskets.

- Can we load these file and create a new attribute “age”?

- Can we find out:
  - how often each customer went shopping
  - how much (s)he bought together (and on average)

Problems with Data and Data Cleaning



- Database issues

- More details regarding pre-processing later:
  - Normalization
  - Binning
  - Feature (and Data!) Reduction
  - …

The 80% Rule
Over 80% of data analysts' time is spent on loading and cleaning data.

Data Understanding



✓ **Goal of the Data Understanding phase**

- Gain general insights about the data that will potentially be helpful for the further steps in the data analysis process

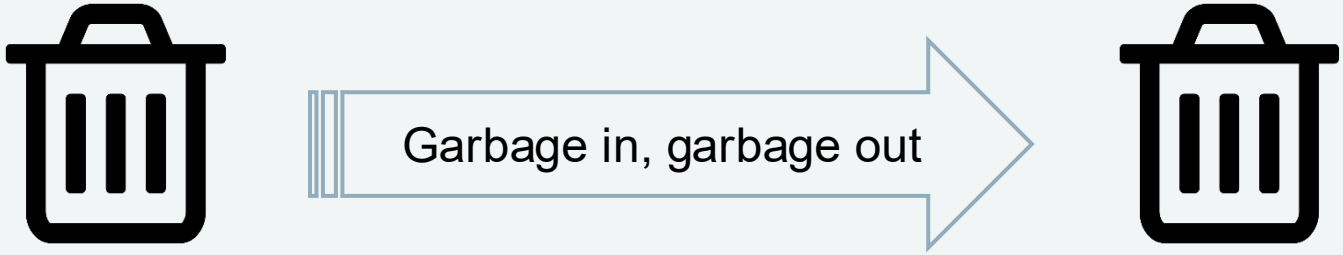
✓ **Reasons**

- Never trust any data as long as you have not carried out some simple plausibility checks.

✓ **Results**

- At the end of the data understanding phase, we know much better whether the assumptions we made during the project understanding phase concerning representativeness, informativeness, data quality, and the presence or absence of external factors are justified

Data Quality



- Data quality refers to how well the data fit their intended use
- There are various data quality dimensions
  - Accuracy
  - Completeness
  - Unbalanced Data
  - Timeliness

Accuracy



Accuracy is defined as the closeness between the value in the data and the true value.

Syntactic

- The value does not belong to the domain of the corresponding
- Easy to spot: verify values lying in the domain

e.g. “fmale” for the attribute Gender and “-15” for the attribute Weight violate the syntactic accuracy

Semantic

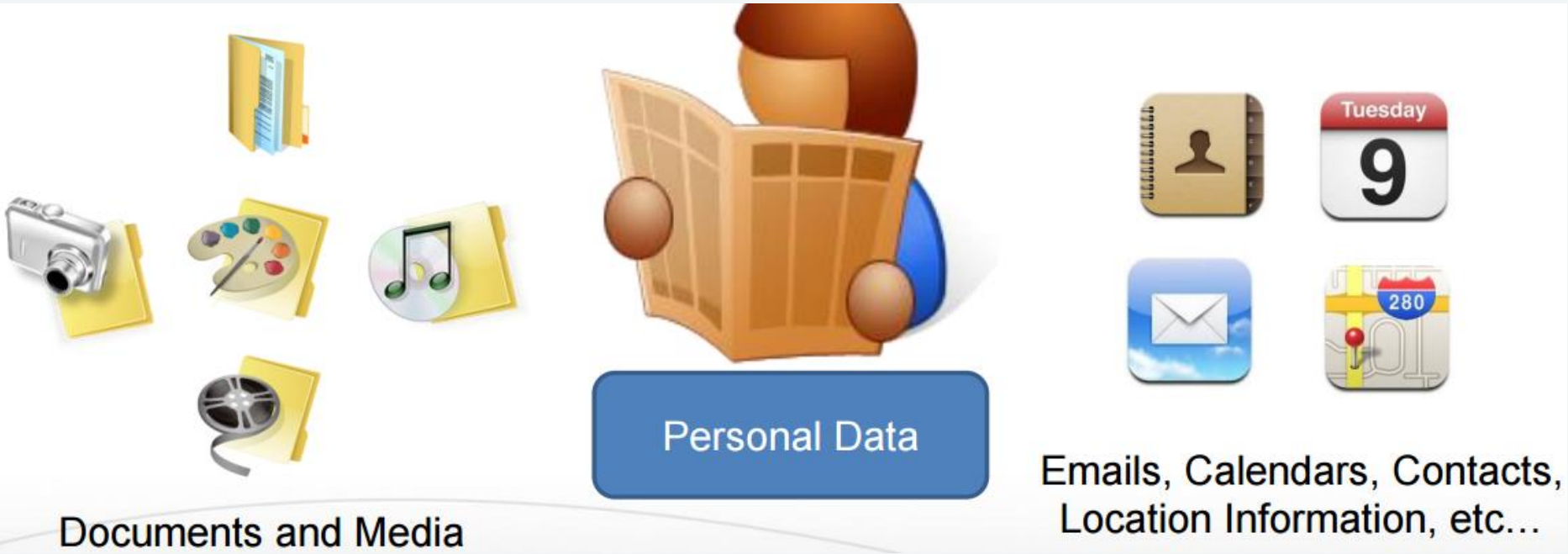
- The value might be in the domain of the corresponding attribute, but it is not correct
- Hard or impossible to spot: double check with other sources or check “business rules”

e.g. “*2090*” for the attribute *YearOfBirth* is (at least at the moment) surely incorrect, therefore violates the semantic accuracy

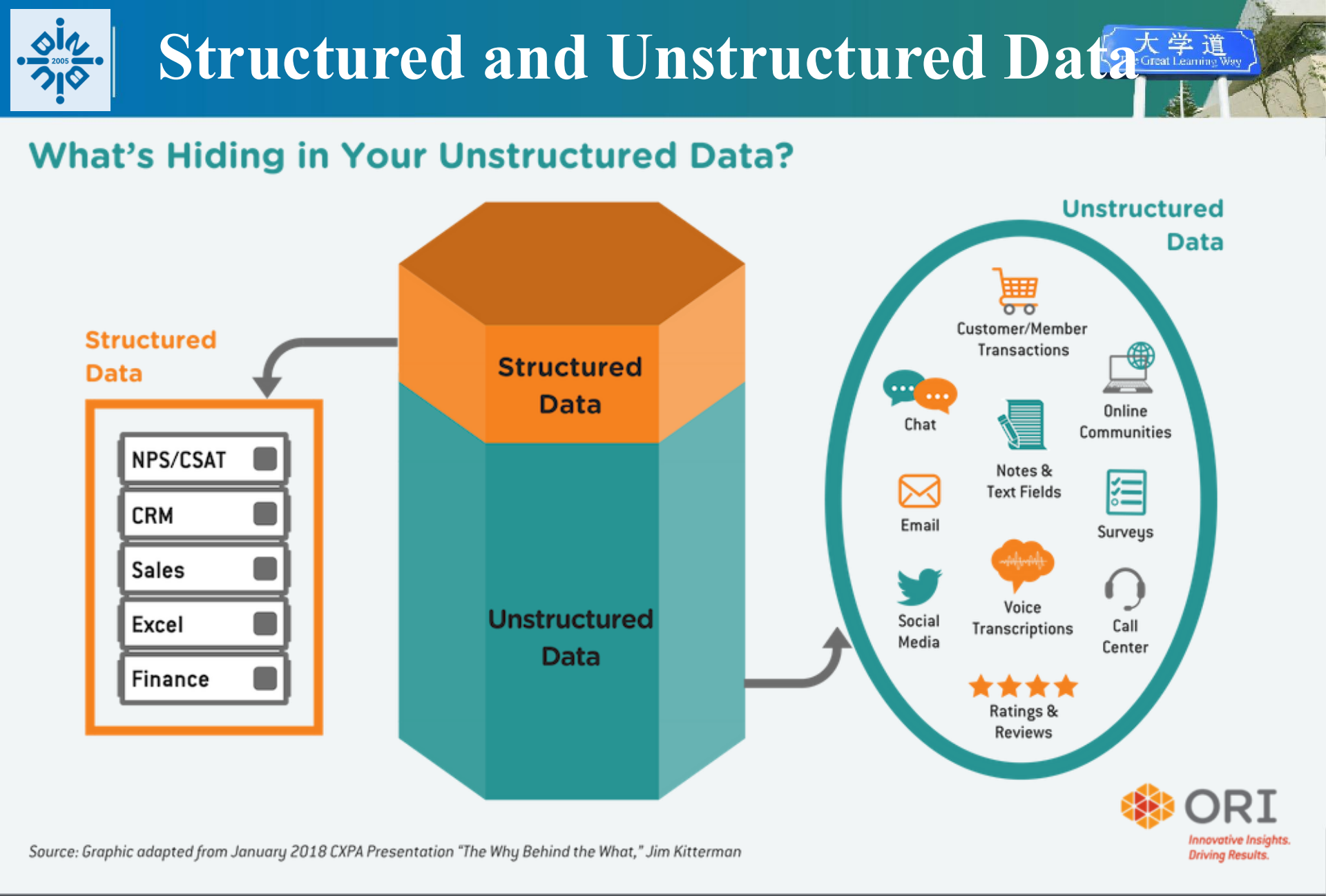
Contents

- Background
- Data Analytics Project Lifecycle
  - Example of Data Analytics Project
  - Project Understanding Phase
  - Data Understanding Phase
- **Data Format**
- Data Types
- Lab: Workflow in KNIME (Data Import Export)
- Lab: Excel and CSV tables
- Review Questions
- APPENDIX: Introduction to KNIME

We Live in a World of Data…



Structured and Unstructured Data



Source: Graphic adapted from January 2018 CXPA Presentation "The Why Behind the What," Jim Kitterman

From Unstructured to Structured (Example 1)

Data item	Structured*	Unstructured
Time	dd/mm/yyyy 14 September 2017 Prescription start date: dd/mm/yy	'Earlier today Mr X experienced chest pain' 'Operation scheduled on Tuesday'.
Symptoms	N242300 Neuropathic pain 1B1B.00 Cannot sleep—insomnia	'…c/o shooting pain in upper right leg during the night, disturbing her sleep'
Diagnosis	C109912 Type 2 diabetes without complication	'The patient has diabetes without complications'
Prescription	01040200 (BNF code for codeine phosphate 60 mg tablets)	'Px codeine 60 mg PO qid×7 days'
Referral	8H4D.00 Referral to psychogeriatrician	Rev 4 w ?refer pyscho ger
Test	43F1.00 Rheumatoid factor positive	Rheumatoid factor was 42 IU/mL which is a positive result

*These codes represent Read codes, a UK-based, alphanumeric clinical coding system for general practice, and British National Formulary (BNF) codes, which represent the full list of medications available in the UK.

From Unstructured to Structured (Example 2)



- **“Thomas said: For our course, we’ll have our final exam on 9:30, 6/December. The exam will last 2 hours, place will be T7-501. Please revise carefully.”**
- {
  ”date”: 06/12/2021
  “time”: 09:30
  “duration”: 120
  “event”: “final exam”
  “type”: “closed book”
  “place”: “T7-501”
  “course code”: “GFQR1013”
  “course name”: “Hands on Data Analytics”
  “lecturer”: “Thomas Canhao XU”
  }

Common Structured Data Types



- Comma-separated values (CSV)
- Extensible Markup Language (XML)
- JavaScript Object Notation (JSON)
- Microsoft Excel (XLS)

Comma-separated Values



A comma-separated values (CSV) file is a delimited text file that uses a comma to separate values.

- Each line of the file is a data record.
- Each record consists of one or more attributes. The attributes are separated by commas

- A CSV file typically stores tabular data (numbers and text) in plain text, in which case each line will have the same number of fields.

Comma-separated Values



- Adjacent fields must be separated by a single comma.
- However, "CSV" formats vary greatly in this choice of separator character.
- In particular, in locales where the comma is used as a decimal separator, semicolon, TAB, or other characters are used instead.
  - `1997,Ford,E350`

Comma-separated Values



- Any field may be quoted (that is, enclosed within double-quote characters), while some fields must be quoted, as specified in the following rules and examples:
  - **`"1997","Ford","E350"`**
- Fields with embedded commas or double-quote characters must be quoted.
  - **`1997,Ford,E350,"Super, luxurious truck”`**
- Each of the embedded double-quote characters must be represented by a pair of double-quote characters.
  - **`1997,Ford,E350,"Super, ""luxurious"" truck"`**

Example of Quotes in CSV




```
Year,Make,Model,Description,Price
1997,Ford,E350,"ac, abs, moon",3000.00
1999,Chevy,"Venture ""Extended
Edition""","",4900.00
1999,Chevy,"Venture ""Extended Edition,
Very Large""",,5000.00
1996,Jeep,Grand Cherokee,"MUST SELL!
air, moon roof, loaded",4799.00
```

Example of Quotes in CSV



Year	Make	Model	Description	Price
1997	Ford	E350	ac, abs, moon	3000.00
1999	Chevy	Venture "Extended Edition"		4900.00
1999	Chevy	Venture "Extended Edition, Very Large"		5000.00
1996	Jeep	Grand Cherokee	MUST SELL! air, moon roof, loaded	4799.00

Comma-separated Values



- The CSV file format is supported by almost all spreadsheets and database management systems.
- Microsoft Excel also supports CSV.
- CSV format is supported by libraries available for most programming languages.
- Most provide some way to specify the field delimiter, decimal separator, character encoding, quoting conventions, date format, etc.
- **Put it simple, CSV is one of the most widely used data format for structured data.**



Because HTTP is a protocol for transferring *text*, the data you request through a web API needs to be *serialized* into a string format. Often this serialization uses **JavaScript Object Notation (JSON)**. This data format makes their string representations easy to inte

```
{ "title" : "Data Science Book",
  "author" : "Joel Grus",
  "publicationYear" : 2014,
  "topics" : [ "data", "science", "data science"] }
```



Extensible Markup Language (XML) is a markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable.

```
<Book>
  <Title>Data Science Book</Title>
  <Author>Joel Grus</Author>
  <PublicationYear>2014</PublicationYear>
  <Topics>
    <Topic>data</Topic>
    <Topic>science</Topic>
    <Topic>data science</Topic>
  </Topics>
</Book>
```

Example: Unstructured Text Data from the Web




```
The Project Gutenberg eBook of Pride and Prejudice, by Jane Austen

This eBook is for the use of anyone anywhere in the United States and
most other parts of the world at no cost and with almost no restrictions
whatsoever. You may copy it, give it away or re-use it under the terms
of the Project Gutenberg License included with this eBook or online at
www.gutenberg.org. If you are not located in the United States, you
will have to check the laws of the country where you are located before
using this eBook.

Title: Pride and Prejudice

Author: Jane Austen

Release Date: June, 1998 [eBook #1342]
[Most recently updated: February 10, 2021]

Language: English

Character set encoding: UTF-8

Produced by: Anonymous Volunteers and David W:

*** START OF THE PROJECT GUTENBERG EBOOK PRID

THERE IS AN ILLUSTRATED EDITION OF THIS TITLE
[# 42671 ]

cover



      Pride and Prejudice

      By Jane Austen

        CONTENTS

         Chapter 1

         Chapter 2
```

```
Chapter 1

It is a truth universally acknowledged, that a single man in
possession of a good fortune, must be in want of a wife.

However little known the feelings or views of such a man may be
on his first entering a neighbourhood, this truth is so well
fixed in the minds of the surrounding families, that he is
considered the rightful property of some one or other of their
daughters.

“My dear Mr. Bennet,” said his lady to him one day, “have you
heard that Netherfield Park is let at last?”

Mr. Bennet replied that he had not.

“But it is,” returned she; “for Mrs. Long has just been here, and
she told me all about it.”
```

Example: Structured Text Data in JSON




```json
{
  "books": {
    "Jane Austen": [
      {
        "id": 1234567890,
        "source": "gutenberg",
        "title": "Pride and Prejudice",
        "author": "",
        "release date": "01/06/1998",
        "update date": "10/02/2021",
        "language": "English",
        "chapter": 1,
        "paragraph": 1,
        "content": "It is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want of a wife."
      },
      {
        "id": 1234567891,
        "source": "gutenberg",
        "title": "Pride and Prejudice",
        "author": "Jane Austen",
        "release date": "01/06/1998",
        "update date": "10/02/2021",
        "language": "English",
        "chapter": 1,
        "paragraph": 2,
        "content": "However little known the feelings or views of such a man may be on his first entering a neighbourhood, this truth is so well fixed in the minds of the surrounding families, that he is considered the rightful property of some one or other of their daughters."
      }
    ]
  }
}
```

Contents

- Background
- Data Analytics Project Lifecycle
  - Example of Data Analytics Project
  - Project Understanding Phase
  - Data Understanding Phase
- Data Format
- **Data Types**
- Lab: Workflow in KNIME (Data Import Export)
- Lab: Excel and CSV tables
- Review Questions
- APPENDIX: Introduction to KNIME

Data Table: Attribute and Instances



Attributes, features, variables…

No	Sex	Age	Blood pr.	Height	Drug
1	male	20	normal	175,0	A
2	female	73	normal	172,2	B
3	female	37	high	163,8	A
4	male	33	low	171,4	B
5	female	48	high	165,9	A
6	male	29	normal	182,3	A
7	female	52	normal	167,2	B
8	male	42	low	177,2	B
9	male	61	normal	168,4	B
10	female	30	normal	174,9	A

Instances, records, data objects, entries…

- Data can usually be described in terms of table or matrices
- Sometimes data are spread among different table that need to be **joined**

Data Types



Categorical | Ordinal | Numeric

No	Sex	Age	Blood pr.	Height	Drug
1	male	20	normal	175,0	A
2	female	73	normal	172,2	B
3	female	37	high	163,8	A
4	male	33	low	171,4	B
5	female	48	high	165,9	A
6	male	29	normal	182,3	A
7	female	52	normal	167,2	B
8	male	42	low	177,2	B
9	male	61	normal	168,4	B
10	female	30	normal	174,9	A

Numeric | Categorical

- Attributes differ for their **scale type**, according to the type of values that they can assume
- Three scale types:
  - Categorical / Nominal
  - Ordinal
  - Numeric

Data Type: Categorical Attributes



Categorical

No	Sex	Age	Blood pr.	Height	Drug
1	male	20	normal	175,0	A
2	female	73	normal	172,2	B
3	female	37	high	163,8	A
4	male	33	low	171,4	B
5	female	48	high	165,9	A
6	male	29	normal	182,3	A
7	female	52	normal	167,2	B
8	male	42	low	177,2	B
9	male	61	normal	168,4	B
10	female	30	normal	174,9	A

Categorical

- Categorical (or Nominal) attributes have a finite set of possible values
- Granularity must be taken into account
  - Hierarchical structure of the categories
  - e.g. shallow subdivision: *food, non-food, drinks…*
  - further subdivision for drinks: *water, beer, wine…*
  - Which level of granularity is appropriate?
- Dynamic Domain
  - Some attributes have a fixed domain (e.g. months)
  - For other attributes the domain can change over time (e.g. the products in a catalogue)
  - Those attributes must be identified and handled

Data Type: Ordinal Attributes



Ordinal

No	Sex	Age	Blood pr.	Height	Drug
1	male	20	normal	175,0	A
2	female	73	normal	172,2	B
3	female	37	high	163,8	A
4	male	33	low	171,4	B
5	female	48	high	165,9	A
6	male	29	normal	182,3	A
7	female	52	normal	167,2	B
8	male	42	low	177,2	B
9	male	61	normal	168,4	B
10	female	30	normal	174,9	A

- Ordinal attributes have an additional linear ordering offered by the domain
- The ordering does not provide the distance between two object
- e.g. for an attribute containing university degrees, we can state that a *Ph.D* is an higher degree than a *M.Sc.* and that this is higher than a *B.Sc.*.

Data Type: Numeric Attributes



Numeric continuous

No	Sex	Age	Blood pr.	Height	Drug
1	male	20	normal	175,0	A
2	female	73	normal	172,2	B
3	female	37	high	163,8	A
4	male	33	low	171,4	B
5	female	48	high	165,9	A
6	male	29	normal	182,3	A
7	female	52	normal	167,2	B
8	male	42	low	177,2	B
9	male	61	normal	168,4	B
10	female	30	normal	174,9	A

Numeric discrete

- The domain of numerical attributes are numbers. They can be
- **Discrete**
  - e.g. age, count…
  - Represented as integer values
- **Continuous**
  - e.g. height, weight, distance…
  - Represented as real values
  - Precision (rounding) has to be handled
- The scale of numeric attributes can be:
  - Interval e.g. date
  - Ratio Scale e.g. distance, with a canonical zero value
  - Absolute Scale e.g. counting

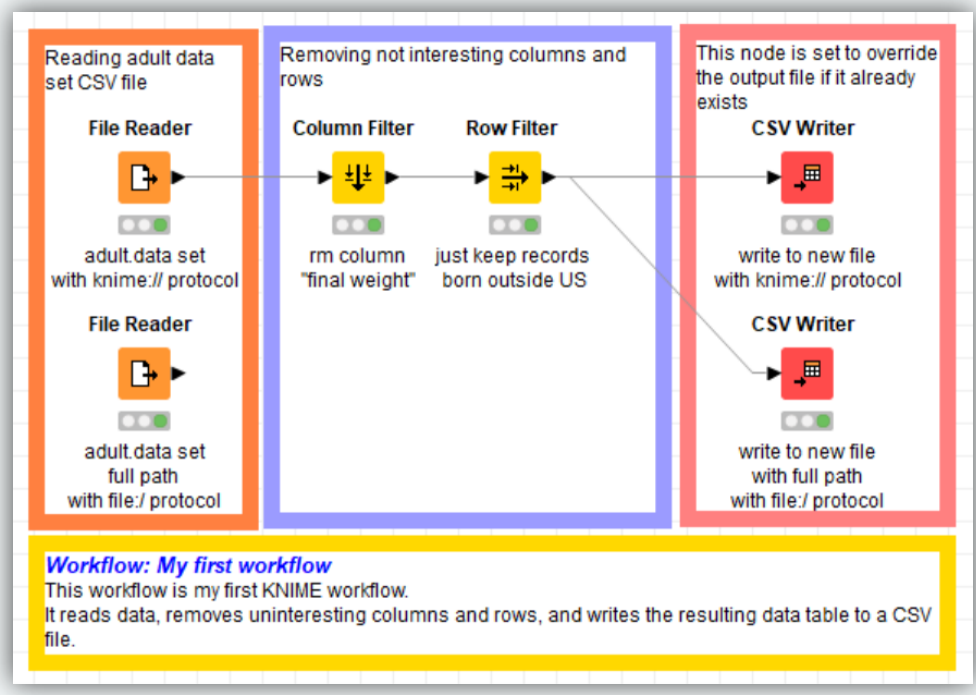
Contents

- Background
- Data Analytics Project Lifecycle
  - Example of Data Analytics Project
  - Project Understanding Phase
  - Data Understanding Phase
- Data Format
- Data Types
- **Lab: Workflow in KNIME (Data Import Export)**
- Lab: Excel and CSV tables
- Review Questions
- APPENDIX: Introduction to KNIME

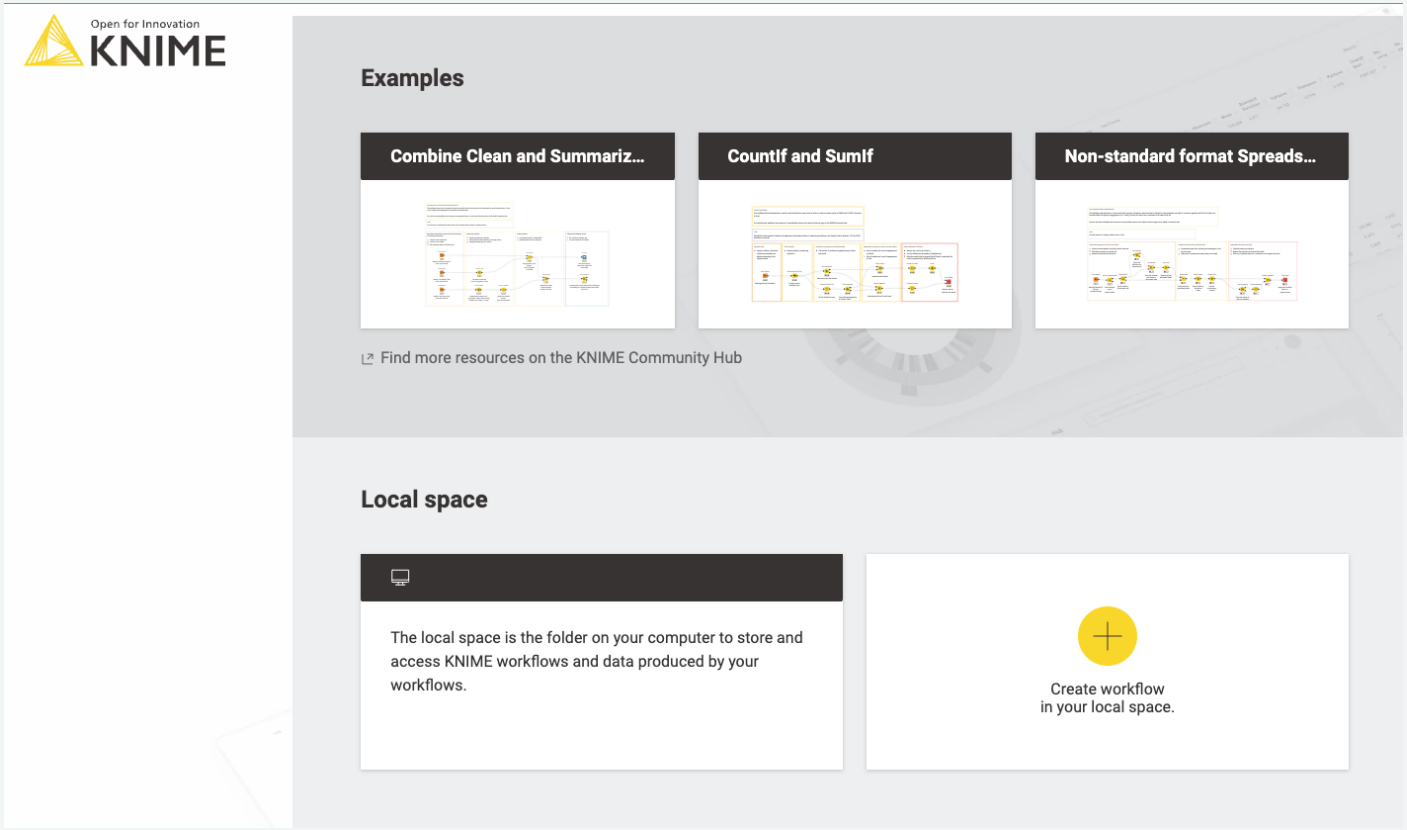
Lab Steps

- Steps
  - Read from CSV or Excel file.
  - Filter rows and columns
  - Write to CSV file

- Dataset used : adult dataset



Creating a New Workflow

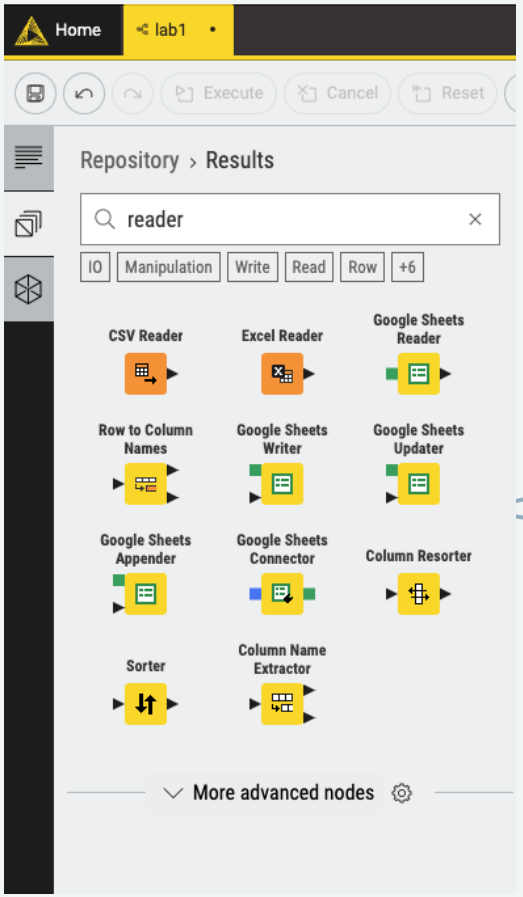


In Home tab of KNIME

-> create a new workflow in your local space

-> name the workflow as "lab1"

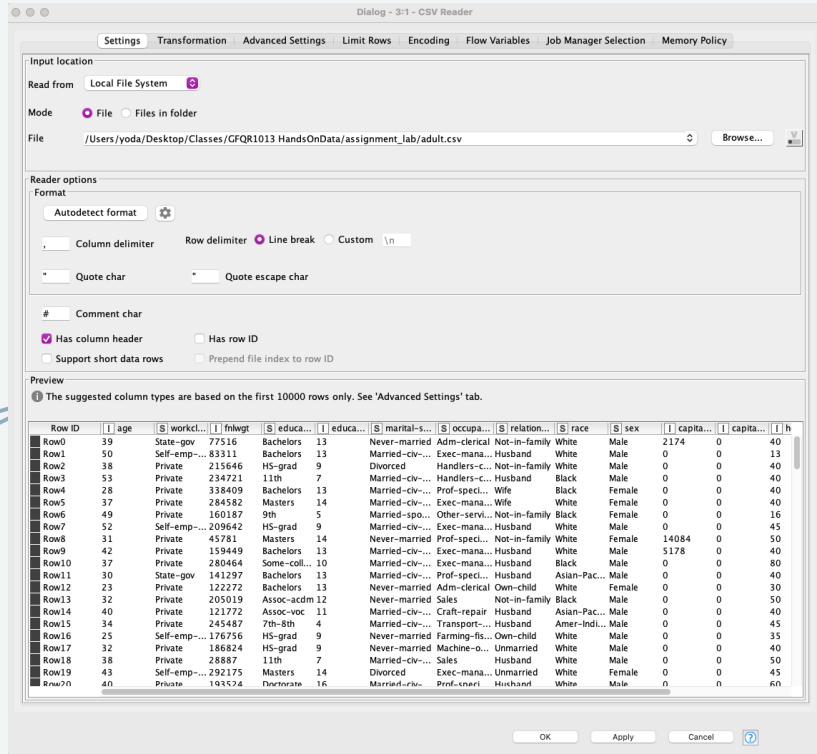
Read the Dataset



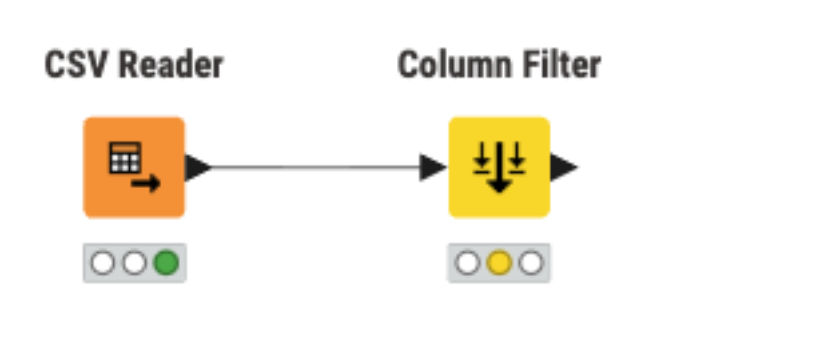


Drag and drop the **CSV Reader** node from the Node Repository to add it to the workflow

Open the configuration window (double click) and select the file on your machine containing the adult dataset

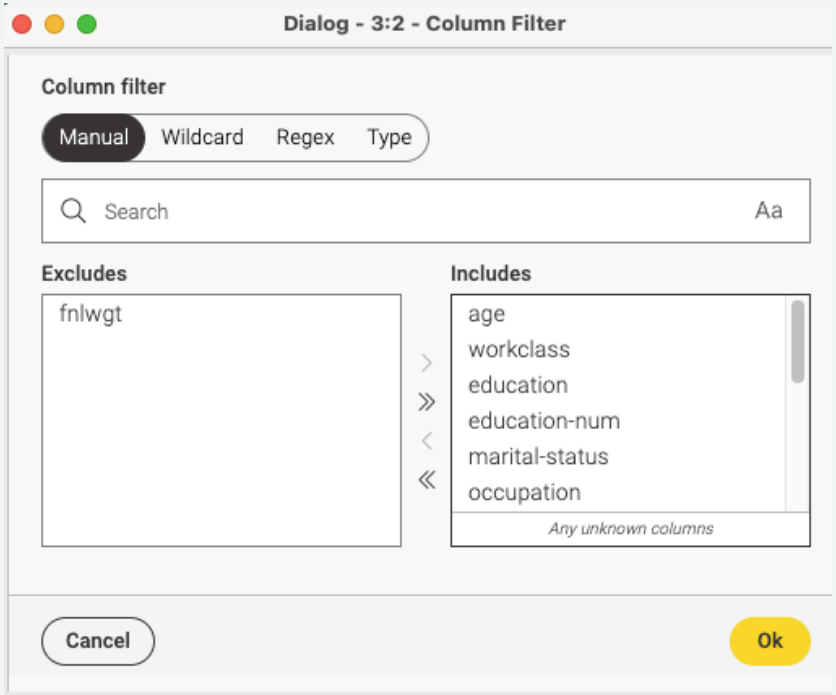


Remove Columns

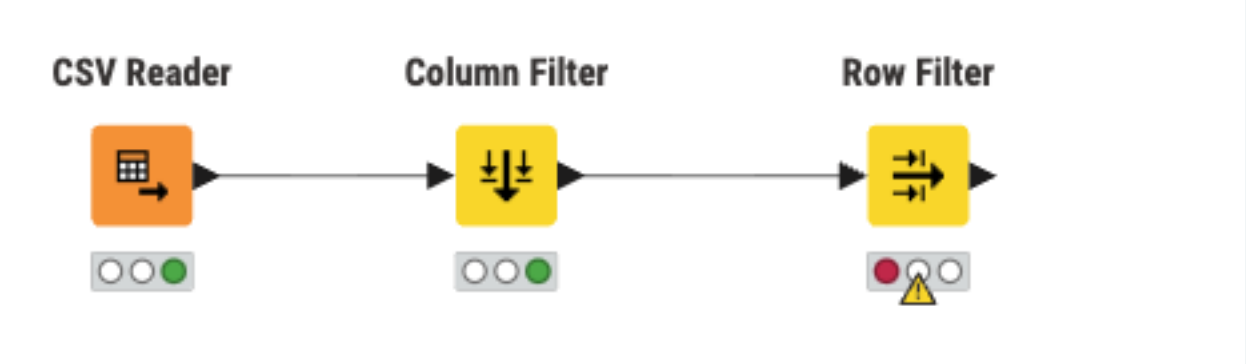


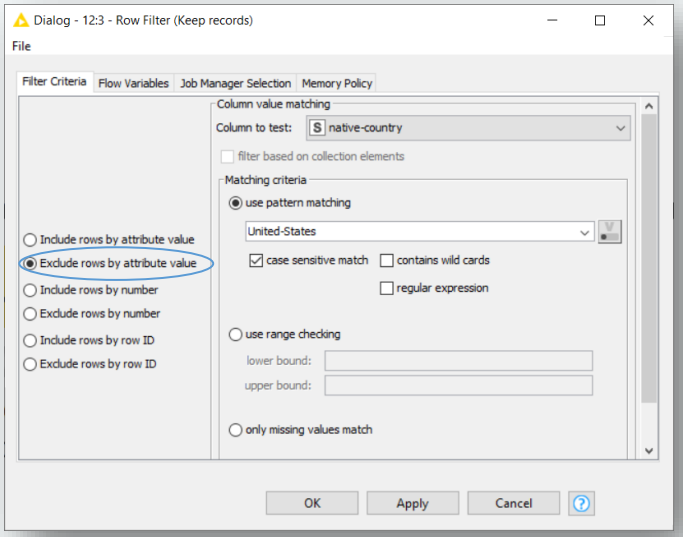
Explore the dataset.

Some columns have unnecessary information. Remove them with a Column Filter node



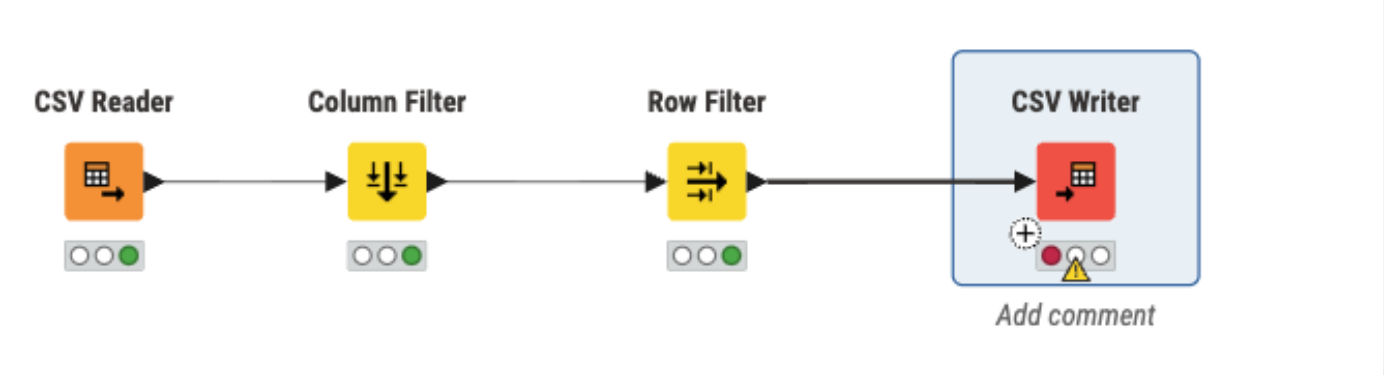
Remove Rows





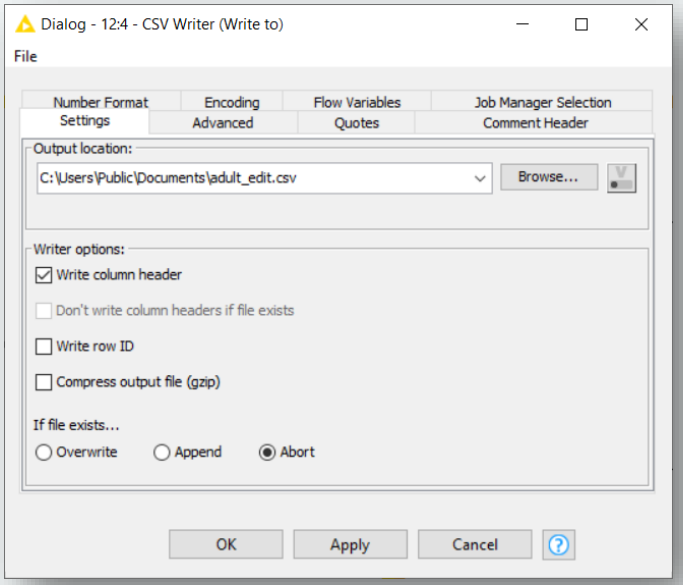
Add a Row Filter node and configure it to only keep entries whose “native-country” value is not “United-States”

Write to a New File



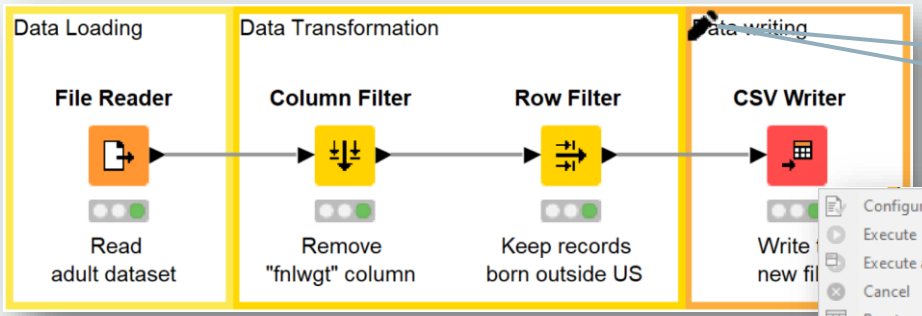
Finally add a CSV Writer node to the pipeline.

Configure and execute it to write the transformed dataset to a new file

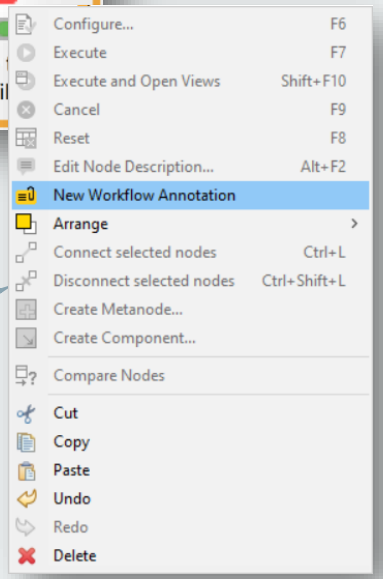


Annotations

- Annotations are coloured editable boxes that you can add to your workflow
- They help you making it more readable and visually pleasant



Double click on the annotation customize its text and appearance



Right click anywhere on your workflow and add a *New Workflow Annotation* from the context menu

Contents

- Background
- Data Analytics Project Lifecycle
  - Example of Data Analytics Project
  - Project Understanding Phase
  - Data Understanding Phase
- Data Format
- Data Types
- Lab: Workflow in KNIME (Data Import Export)
- **Lab: Excel and CSV tables**
- Review Questions
- APPENDIX: Introduction to KNIME

Lab: Excel to CSV



STEPS:

- Open Excel
- Create a data table with 10 random data records with the following columns: **name, year_birth, gender, height_cm, weight_kg**
- Create a new column: **age**
- Create a new column: BMI that is equal to **kg/m^2**
- Save the data to CSV

Contents

- Background
- Data Analytics Project Lifecycle
  - Example of Data Analytics Project
  - Project Understanding Phase
  - Data Understanding Phase
- Data Format
- Data Types
- Lab: Workflow in KNIME (Data Import Export)
- Lab: Excel and CSV tables
- **Review Questions**
- APPENDIX: Introduction to KNIME

Review Questions

Question 1: Provide a sufficient description about the knowledge and skills that one needs to possess to become a successful data analyst.

Question 2: What is structured and unstructured data?

Question 3: What is the relation and difference between computer science, statistics and data analytics?

Review Questions

Multiple Choice Questions (Only one choice is correct)

1. Where can we use data analytics?

(A) Autonomous Driving **(B) Recommender system** **(C) Financial markets prediction** **(D) Predicting temperature** **(E) All of the above are correct**

2. Which of the following is NOT structured data?

(A) CSV file **(B) Email file** **(C) JSON file** **(D) XML file**

3. The following data looks like …

```
{
  "string": "Hi",
  "number": 2.5,
  "boolean": true,
  "null": null,
  "object": { "name": "Kyle", "age": 24 },
  "array": ["Hello", 5, false, null, { "key": "value", "number": 6 }],
  "arrayOfObjects": [
    { "name": "Jerry", "age": 28 },
    { "name": "Sally", "age": 26 }
  ]
}
```

(A) CSV **(B) JSON** **(C) XML** **(D) Excel** **(E) PPT**

Review Questions

4. In ".csv" file format, **new records** are separated by which of the following?

   (A) Comma (B) New line (C) Semicolon (D) Tab (E) Undefined

5. Consider the following table, what is the corresponding CSV format?

2021	Data Science	ZHANG San	Piano, basketball,"laugh"

   (A) 2021, Data Science, ZHANG San, "Piano, basketball, ""laugh"""
   (B) 2021, Data Science, ZHANG San, ""Piano, basketball, ""laugh""""
   (C) 2021, Data Science, ZHANG San, Piano, basketball, "laugh"
   (D) 2021, Data Science, ZHANG San, "Piano, basketball, "laugh""

6. What data type should **Blood Type: {A, B, AB, O}**?

   (A) Categorical (B) Ordinal (C) Text (D) Numerical (discrete) (E) Numerical (continuous)

7. In the context of data types, which of the following is considered a categorical attribute?

   (A) Age (B) Height (C) Weight (D) Gender

Review Questions

8. What is the data type of **Weekday: {Mon, Tue, Wed, Thu, Fri}**?

(A) Categorical | (B) Ordinal | (C) Text | (D) Numerical (discrete) | (E) Numerical (continuous)

9. Consider the following data table, which statement is **not true?**

Sex	Age	Blood	Height	Drug
male	20	normal	175,0	A
female	73	normal	172,2	B
female	37	high	163,8	A
male	33	low	171,4	B
female	48	high	165,9	A
male	29	normal	182,3	A
female	52	normal	167,2	B
male	42	low	177,2	B
male	61	normal	168,4	B

(A) The table has five attributes | (B) The table has nine instances | (C) Sex and Blood are ordinal data | **(D) Age and Height are numeric data** | (E) Drug is categorical data

10. KNIME uses different colors to represent different status of a node. Which of the following statement is **incorrect**?

(A) a red node means the node has not been configured.

(B) A yellow node means the node has been properly configured, but not executed yet

(C) A Green node means the node has been executed successfully

(D) When a node is re-configured properly, its color will turn red again. (Assuming it was green previously)

Contents

- Background
- Data Analytics Project Lifecycle
  - Example of Data Analytics Project
  - Project Understanding Phase
  - Data Understanding Phase
- Data Format
- Data Types
- Lab: Workflow in KNIME (Data Import Export)
- Lab: Excel and CSV tables
- Review Questions
- **APPENDIX: Introduction to KNIME**

KNIME Analytics Platform



- Open and opensource modular Data Science platform
- Covers all the data science needs:

Data Access	Data Preparation	Data Visualization
Machine Learning	Testing	Deployment

Open for Innovation
KNIME

- Based on the visual programming paradigm
- Provides a diverse array of extensions:
  - Text Mining
  - Network Mining
  - Cheminformatics
  - Deep Learning
  - Many integrations, such as Java, R, Python, Weka, Keras, Plotly, H2O, etc
  - ... And more

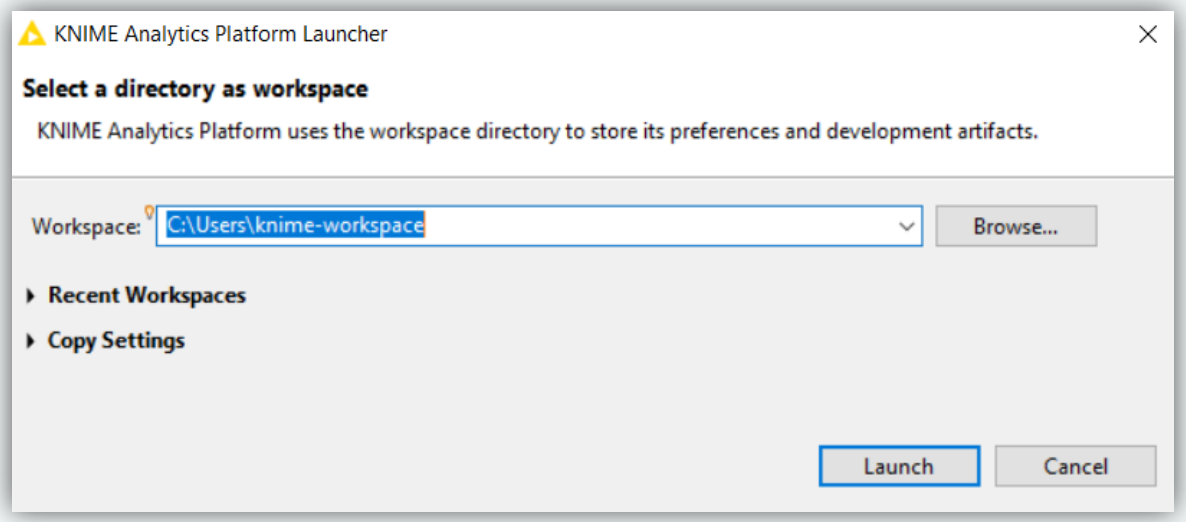
KNIME Installation



https://www.knime.com/downloads

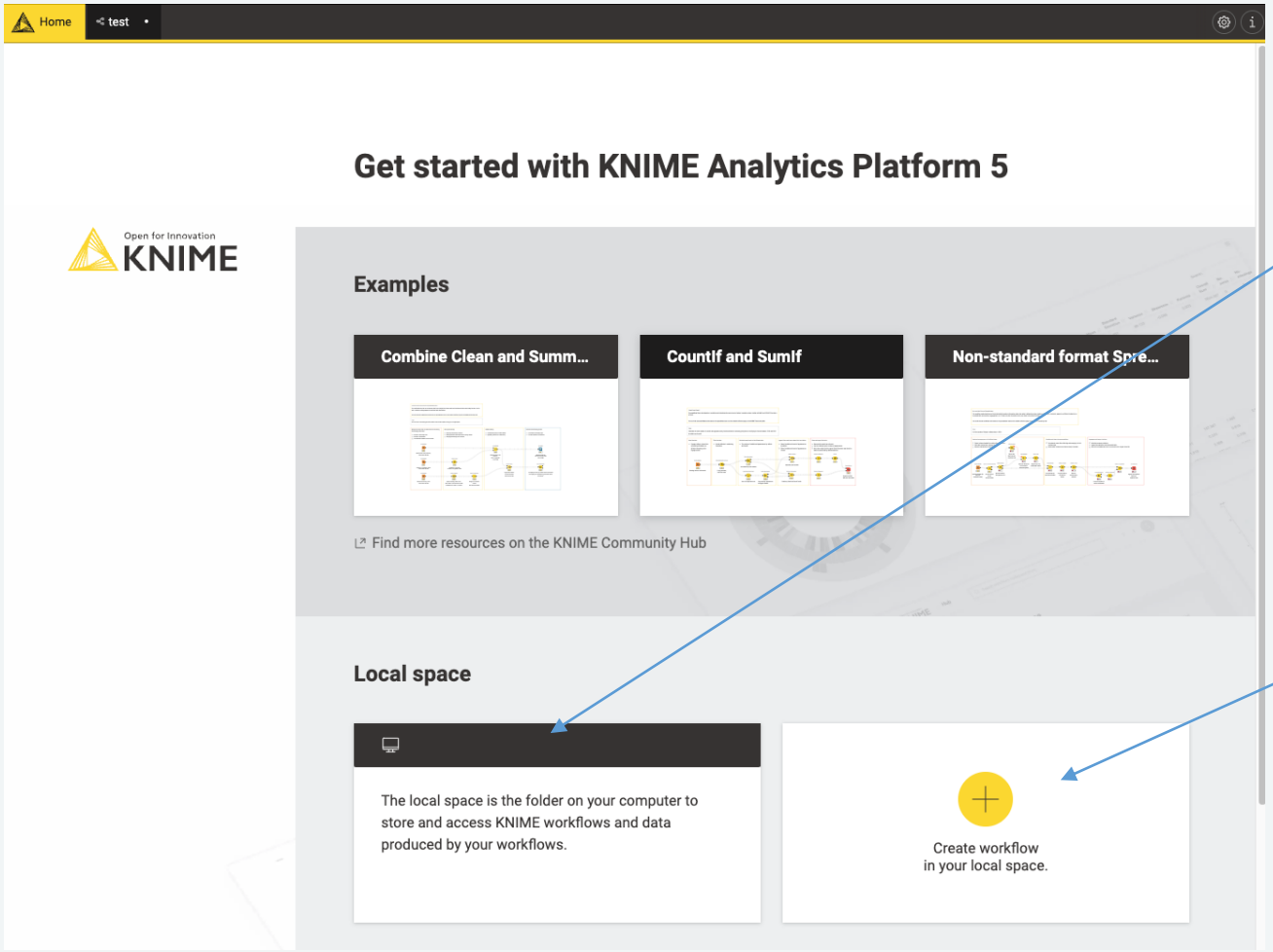
- Select the KNIME Analytics Platform version for your computer:
  - Mac
  - Windows – 32 or 64 bit
  - Linux
- Download the archive and extract the file, or download the installer package and run it

The KNIME Workspace



- The workspace is the **folder/directory** in which workflows (and potentially data files) are stored for the current session.
- Workspaces are portable (just like KNIME Analytics Platform)

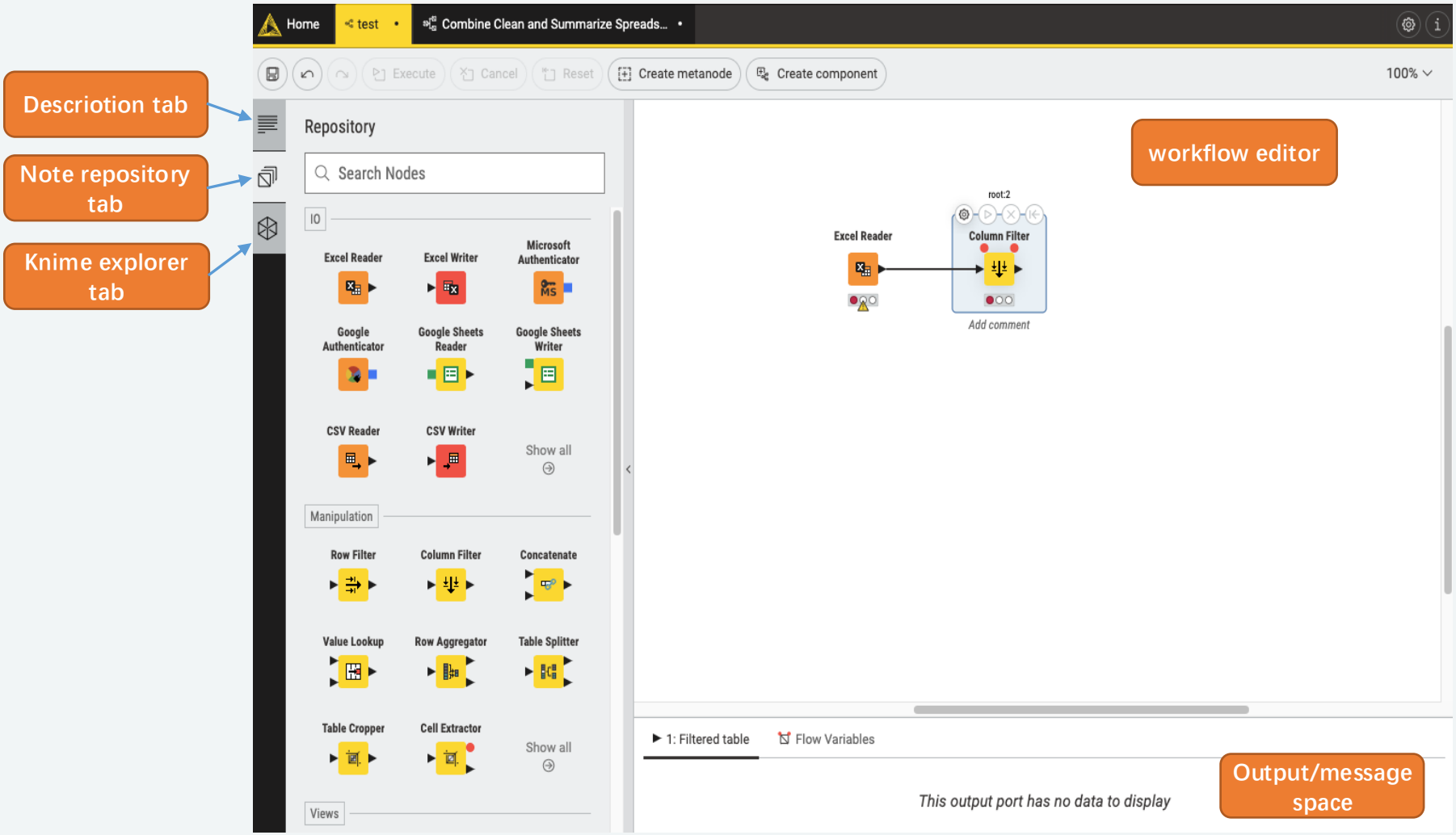
Crate a New Workflow (Project)



Import existing workflow you saved

Create a new workflow

The KNIME Workbench



Importing and Exporting Workflows

Select the "Local space" tab:

To import

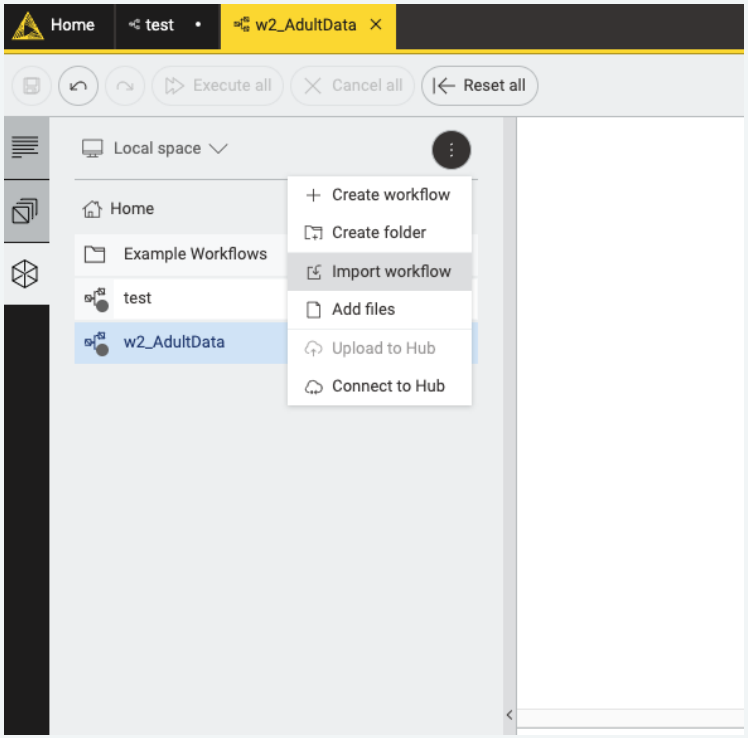
Click on the 3 dots ->

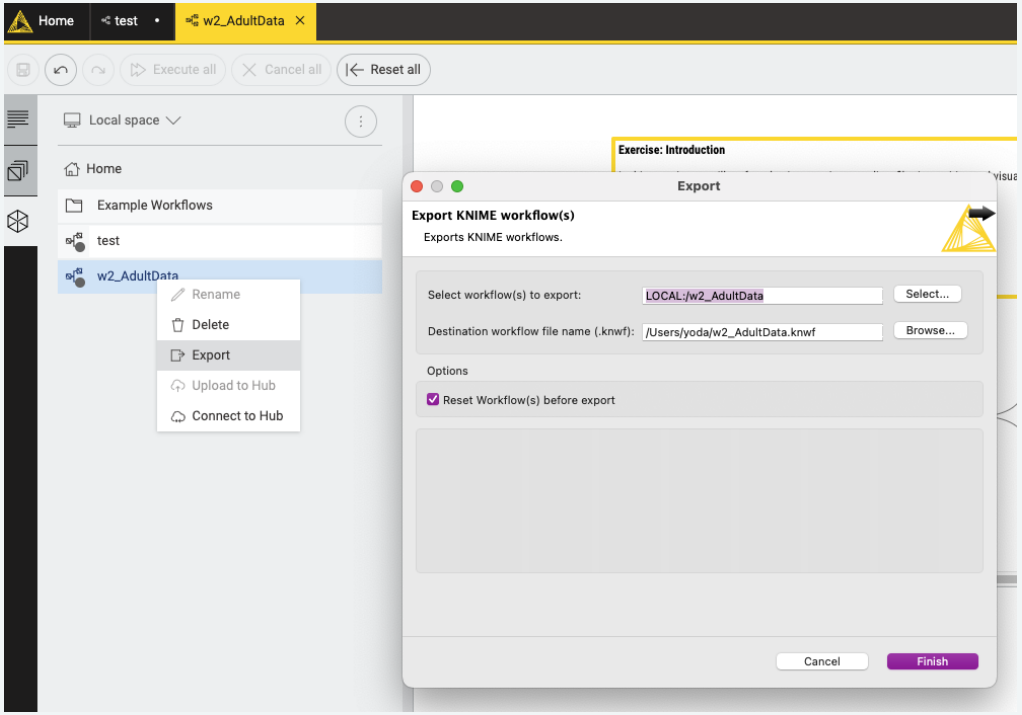
import workflow and select the workflow you saved

To export:

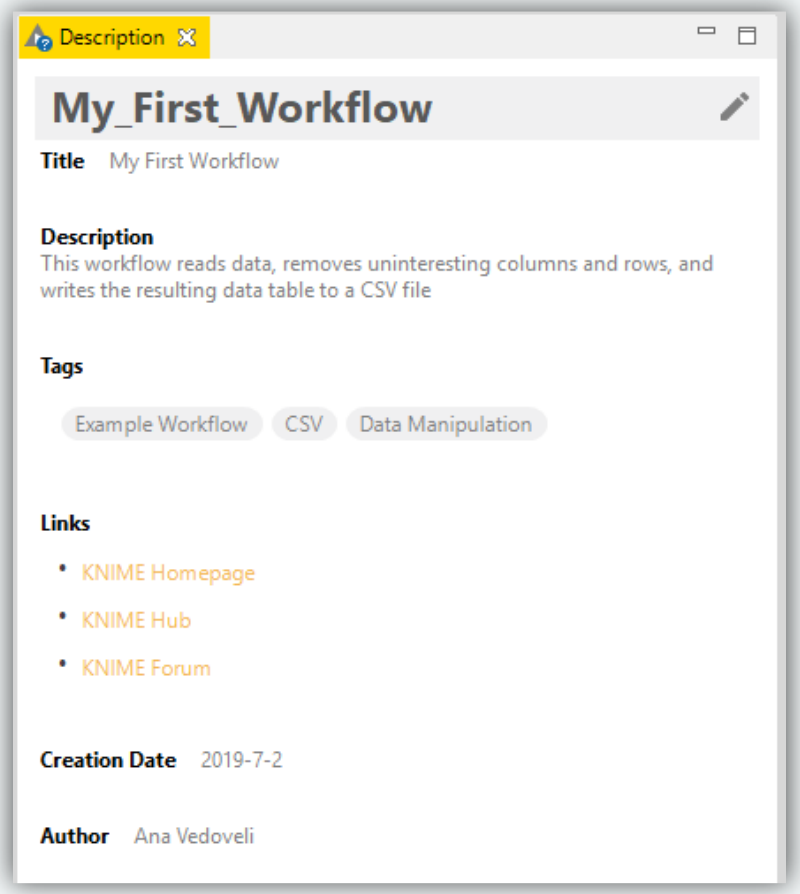
Right-click on a workflow - > export to a destination path

Workflow formats are .knwf (single workflow) and .knar (archives)



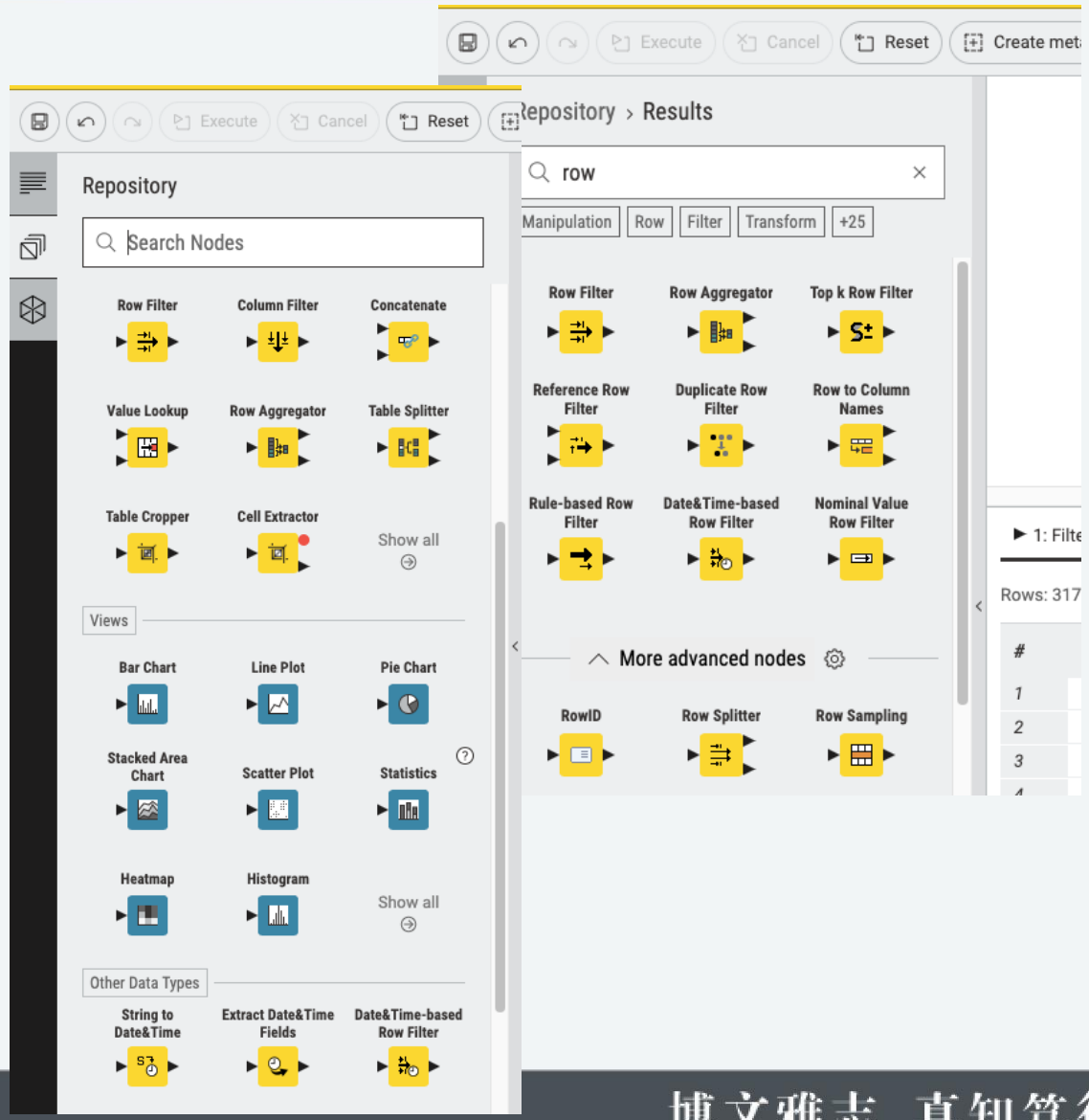


Workflow Description



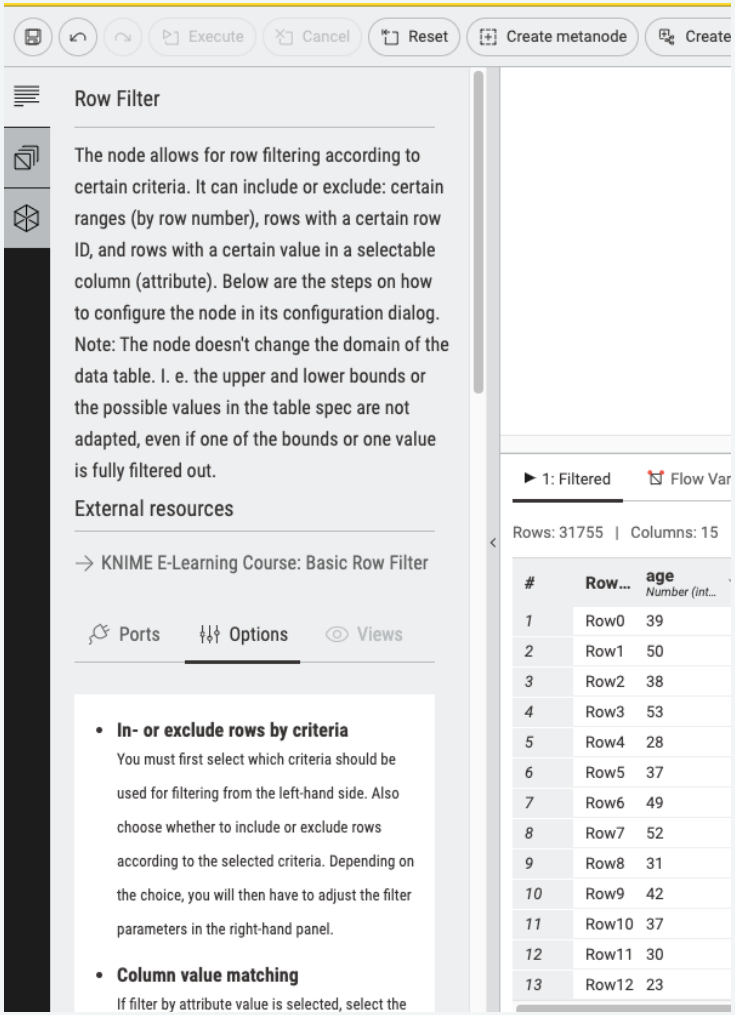
- When selecting the workflow, the Description window gives information about the workflow's:
  - Title
  - Description
  - Associated Tags and Links
  - Creation Date
  - Author

Node Repository



- The Node Repository contains all KNIME nodes - ordered by category
- Each node is a data operation
- Nodes can be searched in the search tab
- Nodes can be added by drag and drop from the Node Repository to the Workflow Editor
- Most nodes are part of KNIME and can be used directly but some may require extra installation

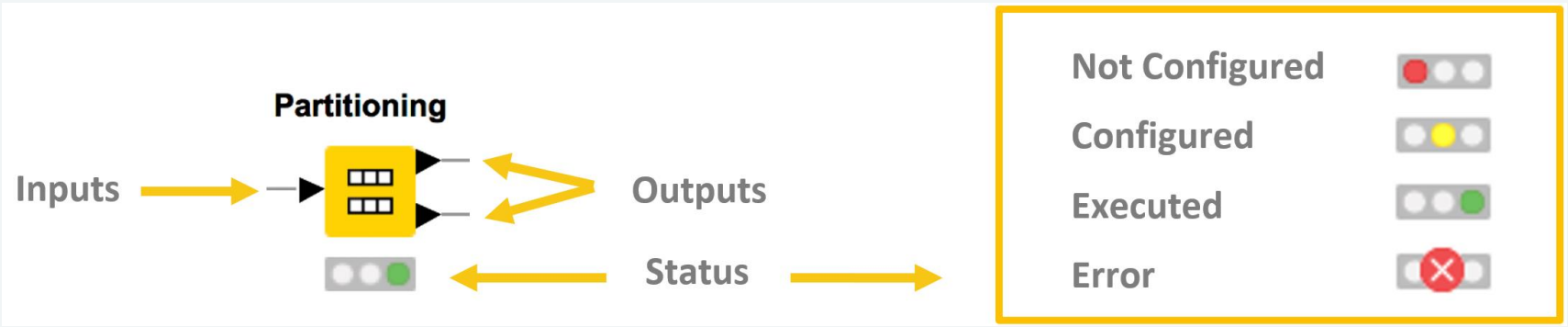
Node Description



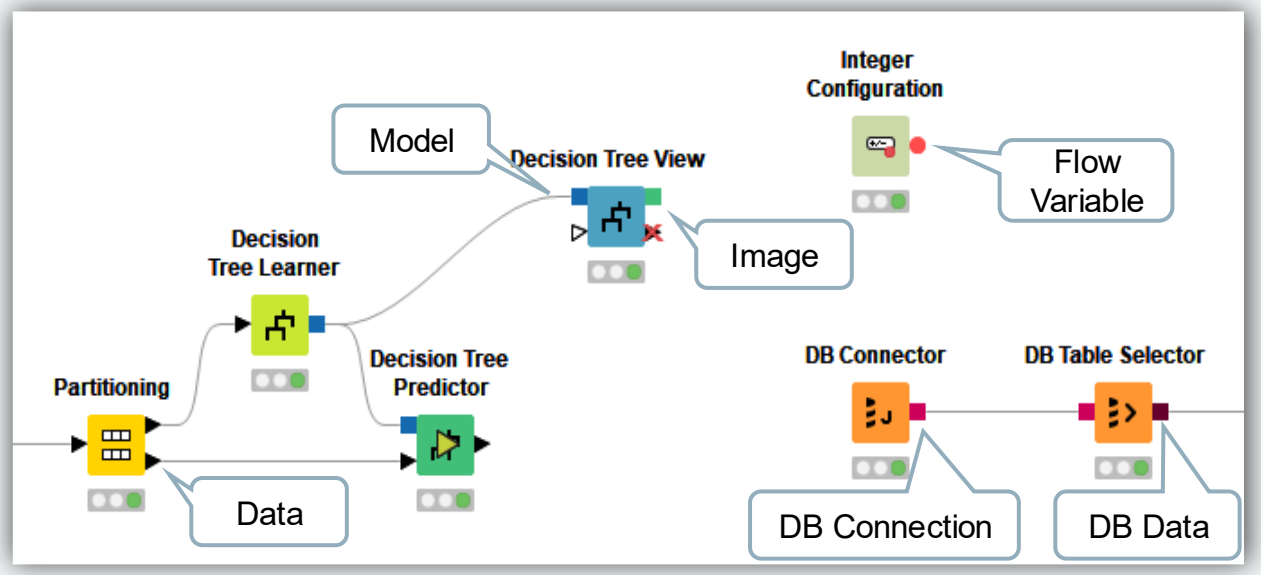
- The Description window gives information about:
  - Node Functionality
  - Input & Output
  - Node Settings
  - Ports
  - References to literature

More on Nodes…

- Nodes are the basic processing units of a workflow
- Each node has a number of input and/or output ports
- Data is transferred over a connection from an out-port to the in-port(s) of other nodes
- Under each node, a light shows its status

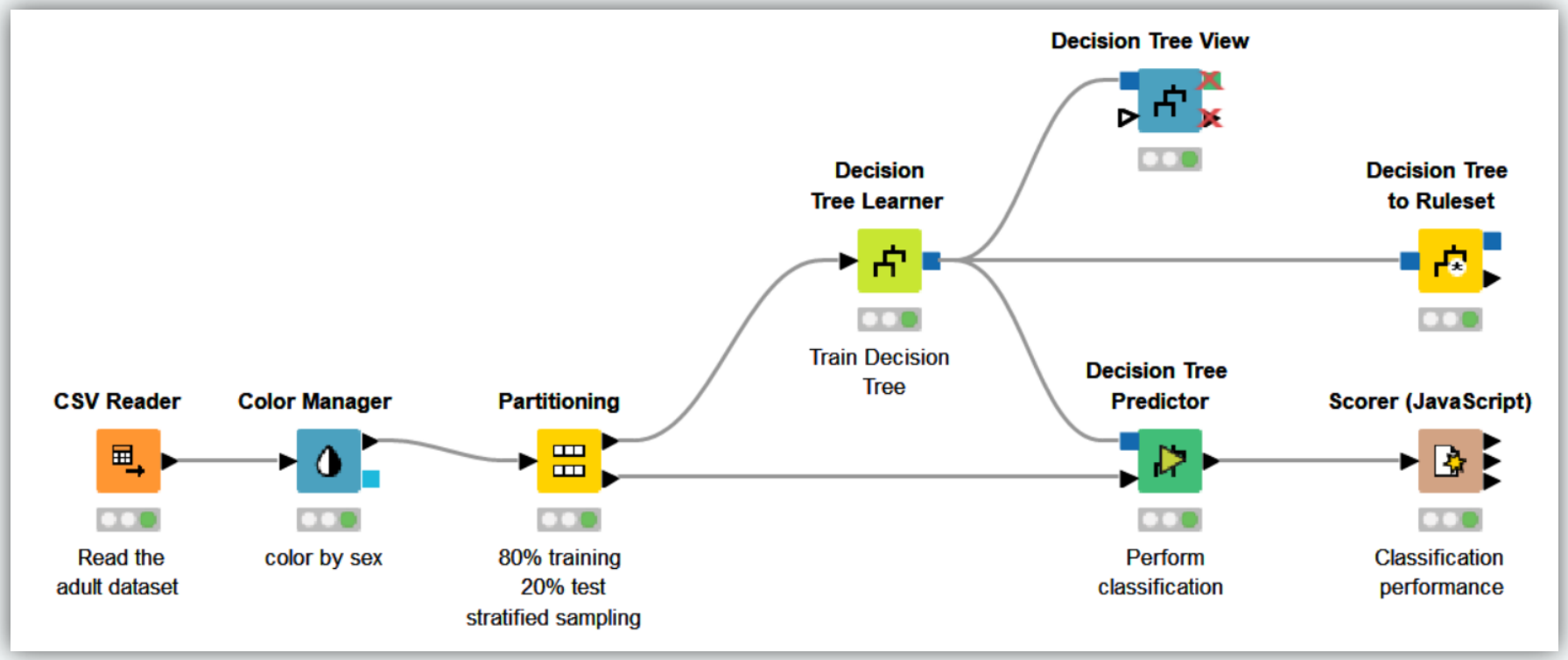


Data Port Types



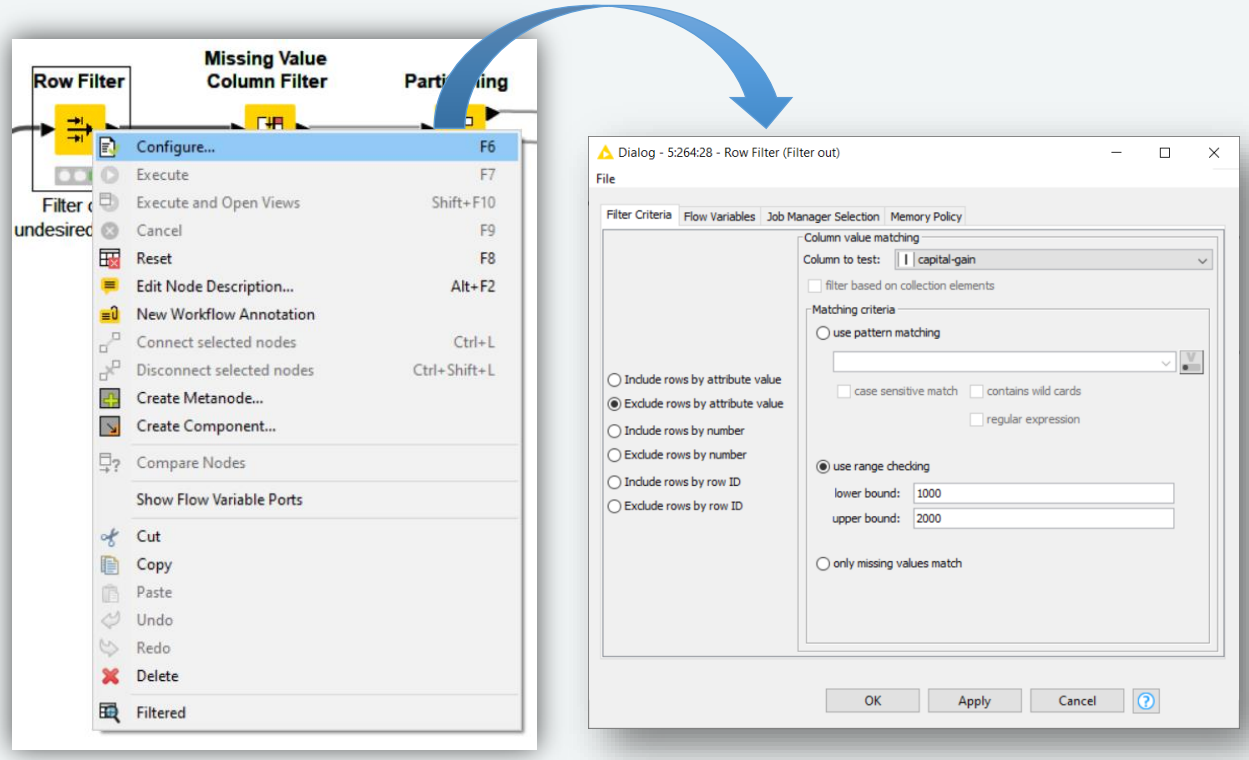
- A pipeline of such nodes makes a **workflow**
- The result of the node's operation on the data is provided at the out-port to successor nodes
- Only port of the same type can be connected

Example of a Workflow



- A workflow is a pipeline of nodes, each configurable to perform a specific task.
  The data flow through nodes from left to right

Node Configuration



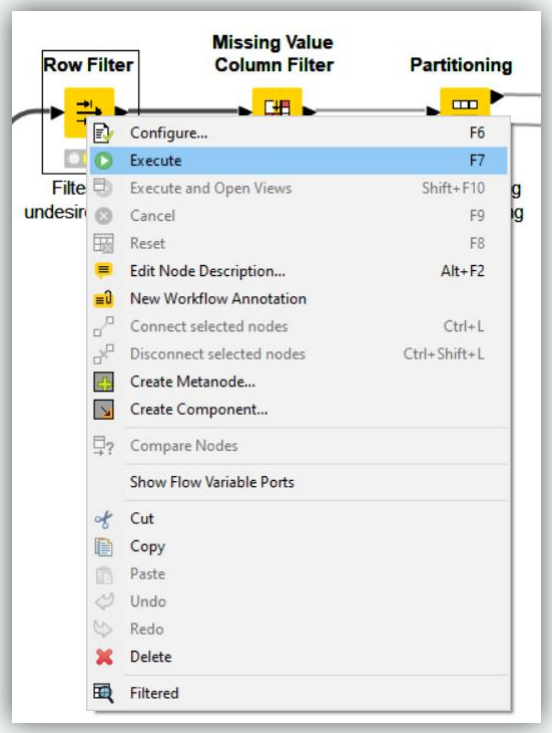
- Most nodes require configuration
- To access a node configuration window:
  - Double-click the node

OR

  - Right-click > Configure

Node Execution

- Right-click node
- Select Execute in context menu



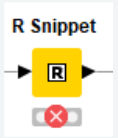
If execution is successful, status shows green light



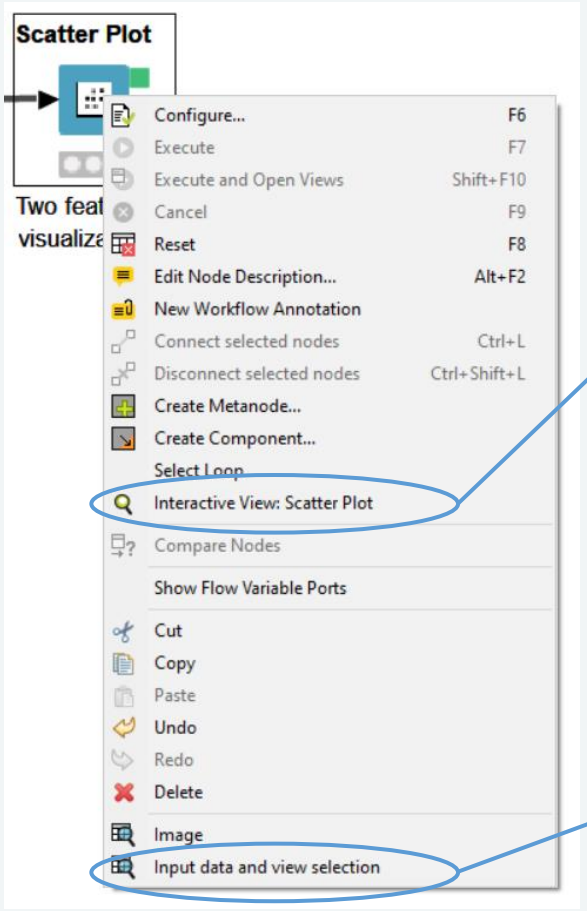
If execution produces warnings, status show yellow triangle

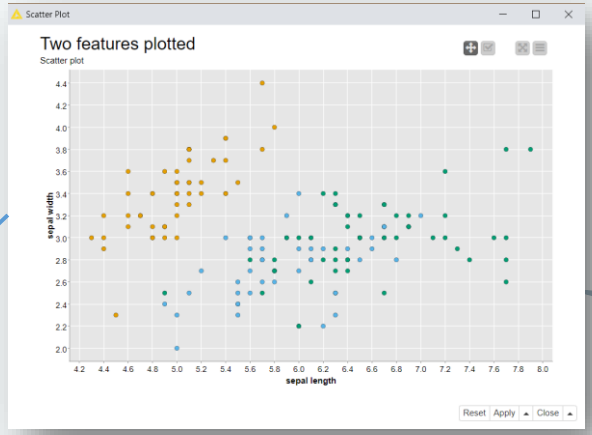


If execution encounters errors, status shows a red X

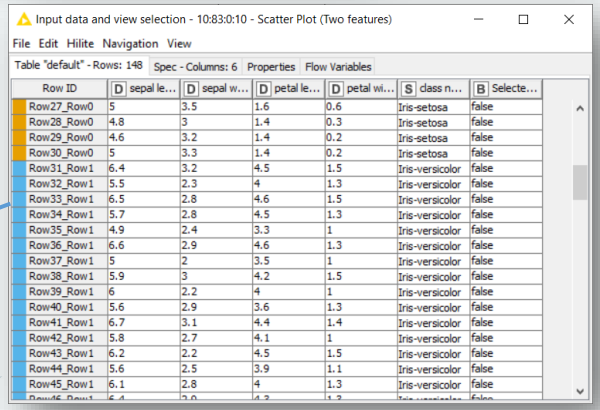


Node Views



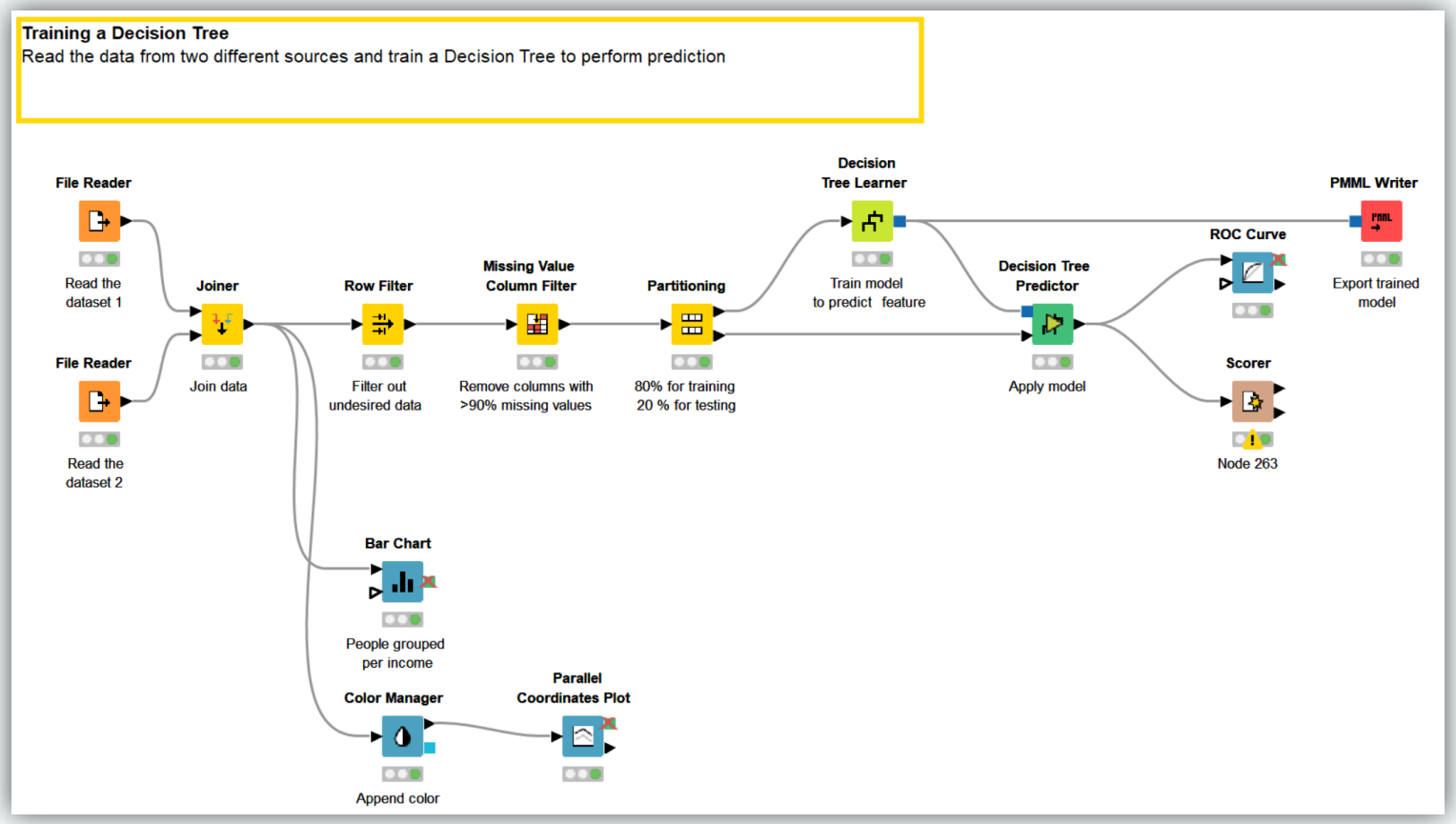


Interactive View



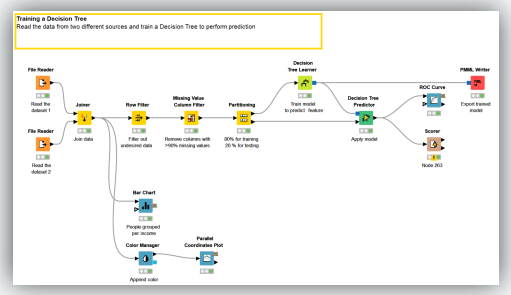
Data View

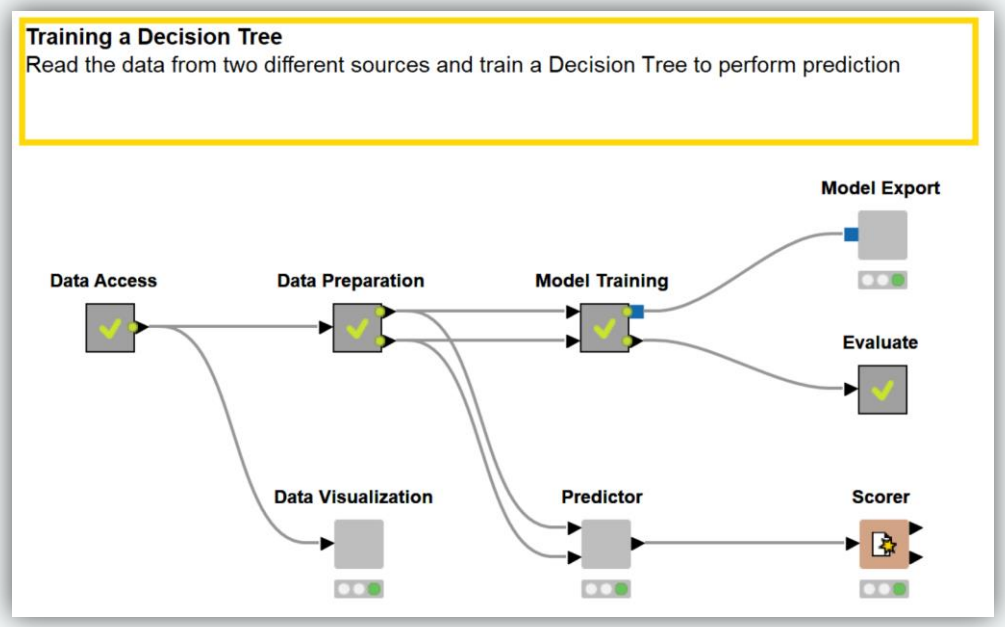
Tidy up workflows



- Workflow can easily become complex and difficult to understand

Tidy up workflows



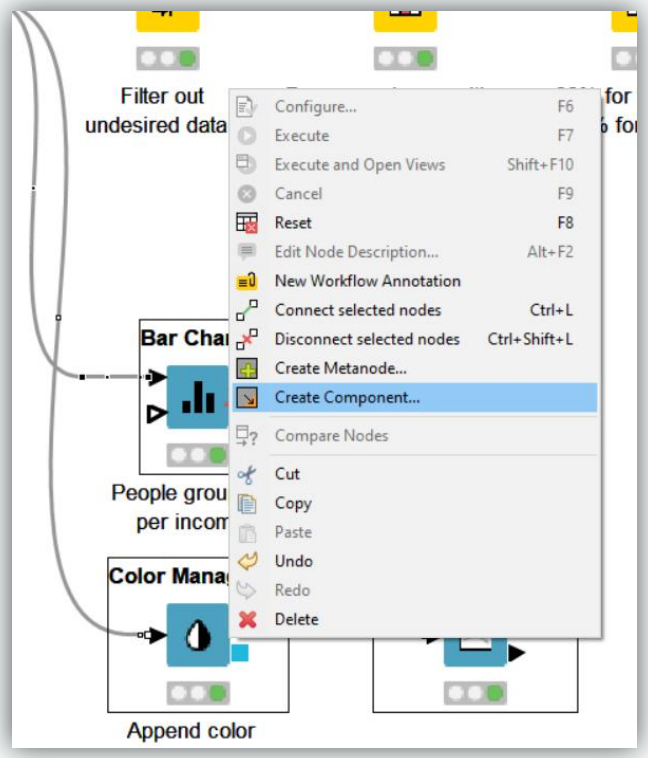


- Metanodes and components can help tidying up, encapsulating nodes performing common operations

Components

Steps to build a component or a metanode

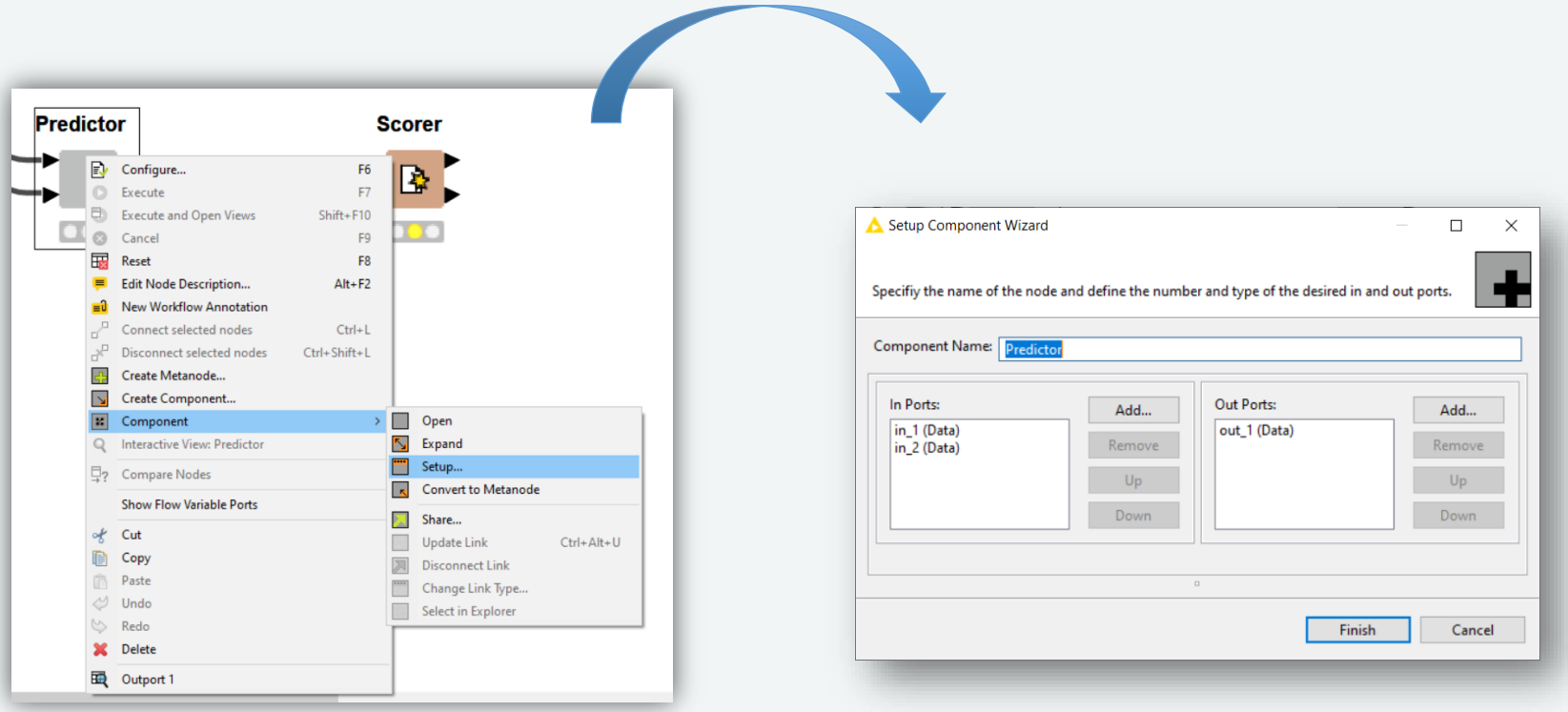
- Select related nodes that you want to group
- Right click
- Select *Create component…* or *Create Metanode…*
- Give it a name



Components have more sophisticated features:

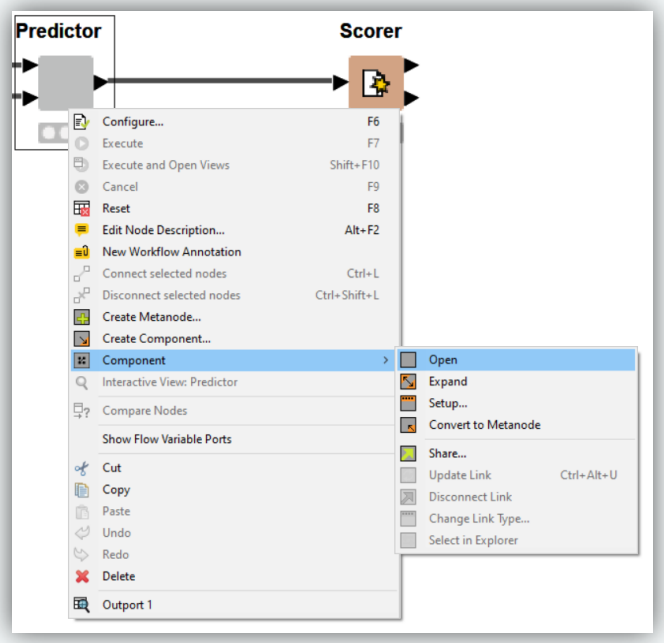
- Encapsulate flow variables, i.e. the parameters only live inside the component
- Provide a **configuration window**: variables and parameters within the component can be edited by Right Click -> Configure…
- Build a **composite view**: Visualization inside the component can be grouped in a dashboard

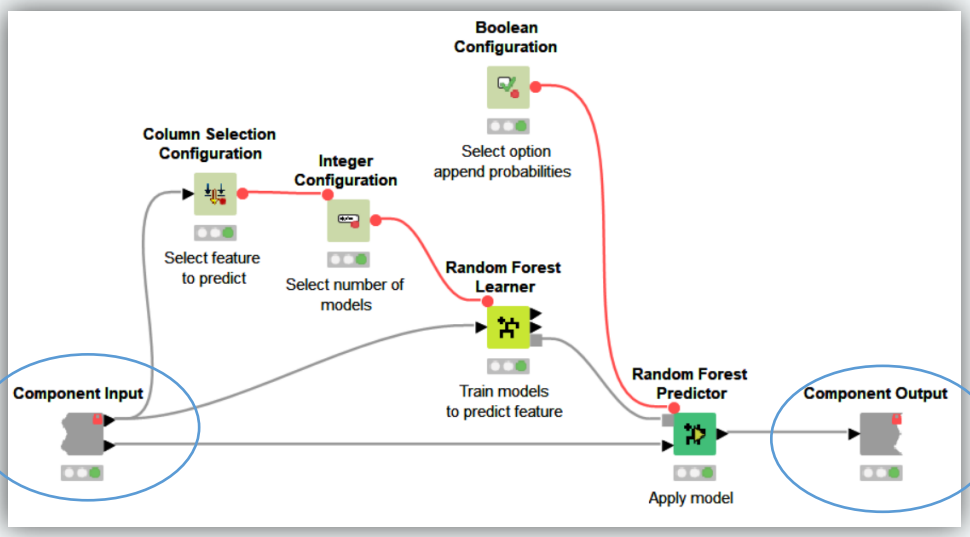
Submenu Component



- Right click on the Component and select Setup··· from the submenu Component to access further customization settings, such as the component name and the ports

Inside a component



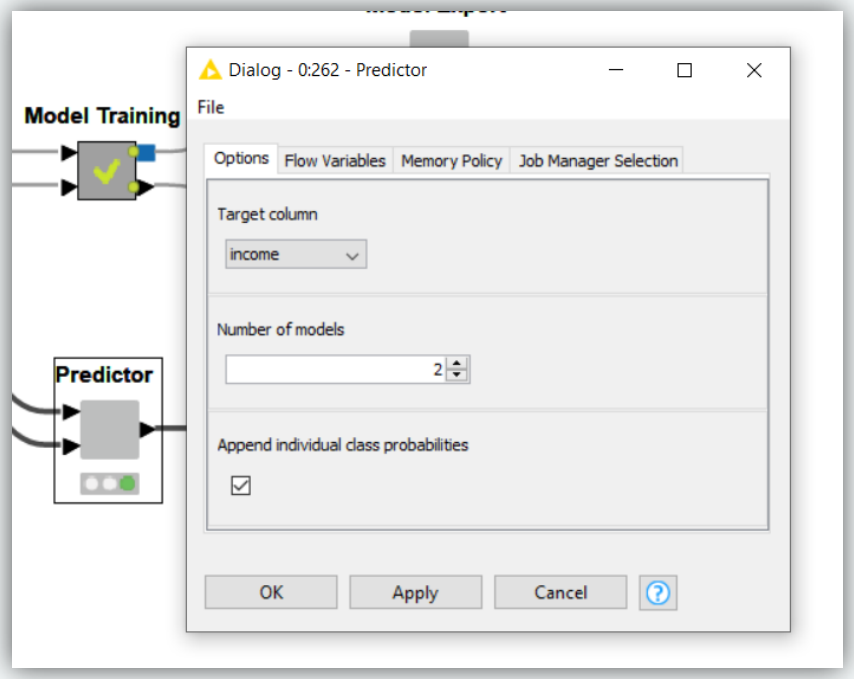


Shortcut:

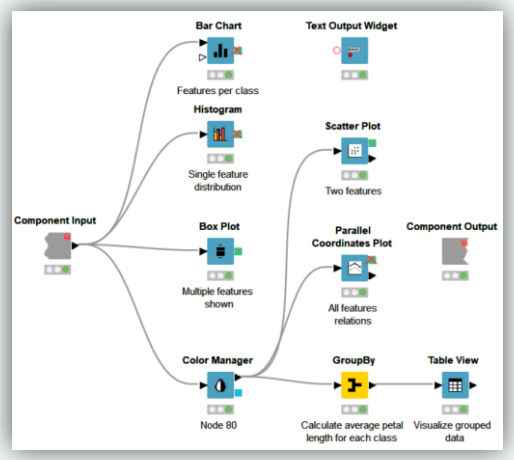
Ctrl + double click on component to open its content

Components Configuration Window

- Components can be configured
- From the configuration window (Right click -> Configure…) the user can enter some parameters
- The entered parameters change the behaviour of the nodes inside the component



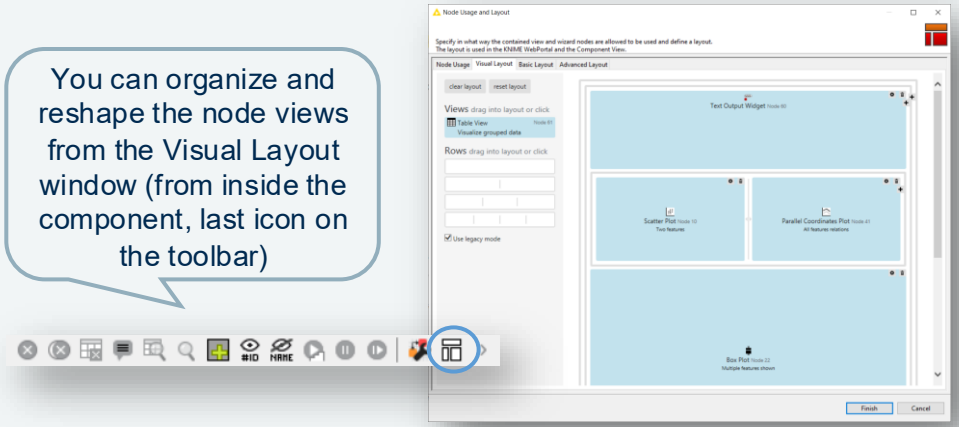
Components Composite View



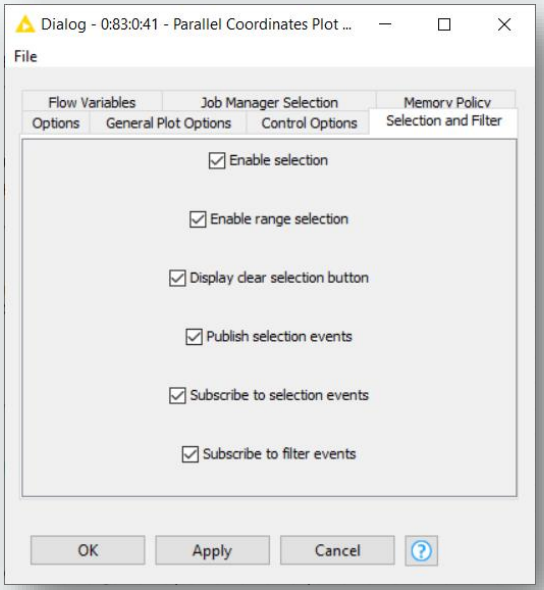
The visualization nodes within the component can be organized to build an interactive composite view

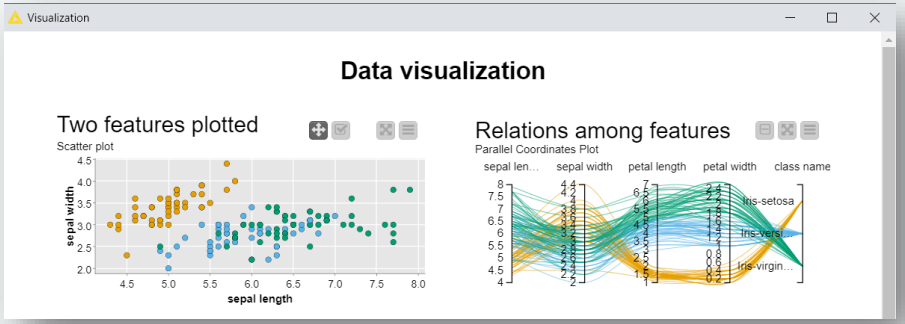


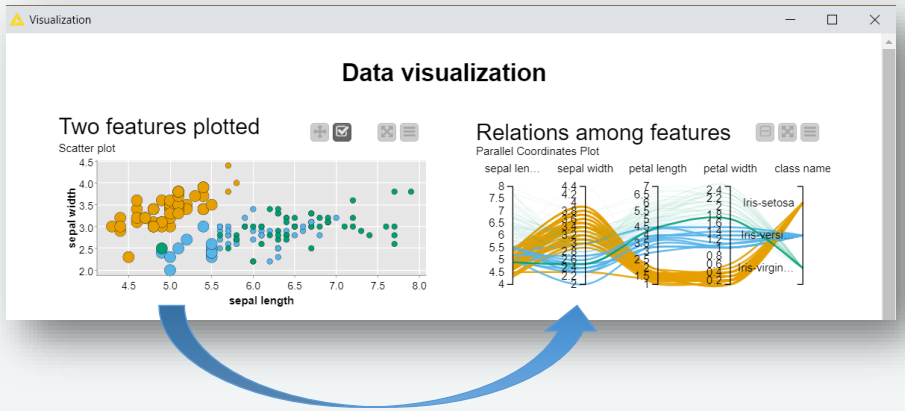
You can organize and reshape the node views from the Visual Layout window (from inside the component, last icon on the toolbar)



Composite views interactivity







Enable publication and subscription to selection events to make the composite view interactive: data selected in one view are highlighted in the others